

An Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA

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TABLE OF CONTENTS

CHAPTER 1: Introduction

The Problem and its Background.....	1
Review of Related Literature.....	4
Significance of the Study.....	31
Statement of the Problem.....	33
Scope and Limitations of the Study.....	35

CHAPTER 2: Methodology

Research Design.....	39
Architectural Framework.....	40
Level 0 DFD/Schematic Diagram.....	42
Algorithm Discussion.....	43
Software Engineering Paradigm.....	58
Population and Sampling.....	61
Data Collection Methods.....	62
Data Analysis Procedures.....	68
Ethical Consideration.....	72

CHAPTER 3: Results and Discussions

Introduction.....	76
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Project Capabilities and Testing Process Details.....	76
Project Testing Results.....	90

CHAPTER 1

INTRODUCTION

The Problem and its Background

An effective resource allocation is one of the important for the success of any organization mostly in project based industries. It involves assigning the right employee to the right task that will be based on their skills and qualifications. The process is still done manually by reviewing internal resumes in many companies. But this method is slow, inconsistent and prone to human error. Organizations grow and will handle more projects, manual resource allocation becomes less efficient and harder to manage.

The technology becomes more essential in improving decision making processes in the increasing amount of data in organizations. According to Roy and Dutta (2022), a recommender system can help the organization to make better decisions by providing accurate suggestions in analyzing data. By adding information, Wang et al. (2023) discuss that using advanced recommendation can improve accuracy of matching people to a specific task or requirements. In the Philippines context, Cacatian et al. (2025) point out that using automation is becoming necessary for companies in Metro Manila to become more competitive and productive.

This study focuses on the WEA, a company that operates in the industrial distribution industry that serves clients in hazardous and highly regulated

environments, including oil refineries, petrochemical plants and other industrial facilities. The company primarily distributes specialized electrical products and equipment designed for hazardous area applications where safety and compliance are critical. The product distribution in WEA provides engineering services such as electrical system design, technical drawings, engineering calculations and project implementation support. WEA is a small industrial electrical distribution company in the Philippines with approximately 14-15 employees. The company is managed remotely by its main office in Singapore. Since WEA does not have dedicated Human Resources (HR) department, employee resource management is currently handled by Finance personnel, while employee project assignments are managed locally by team leads. Because these projects involve complex technical requirements and strict industry standards, employees must possess specific competencies and relevant experience before they can be assigned to a project. With the highly specialized nature of these projects, assigning employees without the appropriate qualifications can lead not only to operational inefficiencies but also to significant safety risks. Therefore, there is a need for a system that can recommend qualified and available employees based on the project requirements.

An Automated Skill-Extraction and Workload-Based Resource Recommendation System is designed to improve workforce allocation at WEA by providing recommendations based on the employee's skill matching, workload, availability and historical performance. The system assists Resource Managers in identifying the most suitable personnel for project assignment while it promotes

balanced resource utilization across the organization. To support these recommendations, the system utilizes OCR and NLP technologies to automate the extraction and processing of the employee skills data from internal resumes. The extracted skills are then validated using a trained Machine Learning classification modal before they are used for recommendation. The classification modal is further improved through user-verified correction, allowing it to continuously enhance its accuracy over time. Aside from the internal workforce allocation, the system also extends to the external hiring process through a Human Resources module, where job openings are posted and applicants upload their resume and manually input their own details, this allow the Human Resources personnel to view the resume alongside the input details to check if they match, before manually filtering and recommending the applicant to supplement the internal talent pool. For the employees, the system will guarantee a fair and transparent selection process, assignment are based on skills, availability and their actual track record rather than manual selection. Once a project is completed, the client's performance rating and feedback are stored and fed back into the system, so that future recommendations are informed by the employee's actual historical performance and not just their skill profile. Finally, this technology shift is the key for WEA to maintain the high project quality and safety standards improving overall productivity in competitive industries market.

Review of Related Literature

Recommender Systems and Fairness in Resource Allocation

Instead of making the recommendations accurate, the modern technology has been shifted in making them also fair and unbiased. According to Jin et al. (2023) and Li et al. (2022), the good system must be designed to give equal chance to all the employees, which is the first need that the study identifies, the need for a fair and unbiased recommendation system that prevents the repeated assignment of the same employees. Building on this, Zhao et al. (2025) further argue that the single focus on utility goals proves to be inadequate in addressing real world concerns, and that the growing attention to fairness aware recommender systems is necessary to truly serve all the users of the system, which identifies the second need of the study, the need for a recommender system that considers both fairness and diversity in its suggestions so that no employee is consistently overlooked or overloaded. This is further supported by Rampisela et al. (2023), who points out that recommendation engines risk reinforcing existing biases by consistently overlooking qualified but underutilized personnel, and with this the system will not recommend only the visible people, leaving others behind, which gives the third need of the study, the need for a provider fairness mechanism that ensures all qualified employees have an equal chance of being recommended regardless of how often they have been selected in the past.

These systems are also essentials for business efficiency. A balanced system must help both the company and employee by distributing the workload more evenly, that are pointed out by Zhou et al. (2024), instead the manager are relying on the gut feeling or memory, a data-driven system approach that actual looks at availability. With this the fourth need of the study is identified, which is the need for a workload tracking system that distributes tasks evenly based on actual availability rather than personal preference or memory of the manager. Zhao et al. (2025) also identify that fairness and diversity in recommender systems are strongly connected, meaning that a system that is fair in its recommendations is also more likely to give diverse and balanced suggestions that will not keep repeating the same candidates, and this gives the fifth need of the study, the need to combine fairness, diversity and workload awareness into one recommendation logic instead of treating them as separate features. These findings gives a strong foundation in addressing the workload problem of this study, showing the only way to truly identify the underutilized talent is through system that can analyzes employee's skills and current tasks without bias.

The literature consistently shows that the core problem of WEA is not just a matter of inefficiency but also of fairness. Jin et al. (2023) and Li et al. (2022) establish that a good recommender system must give equal chance to all employees, while Zhao et al. (2025) strengthen this by arguing that fairness and diversity in recommendations are strongly connected and must both be considered in the system design. Rampisela et al. (2023) further confirm that without a systematic and objective filter, recommendation engines will keep

reinforcing the same biases that WEA is currently experiencing, and the workload problem identified through Zhou et al. (2024) shows that distributing tasks evenly based on actual availability is a necessary condition for a fair system. With all of these five needs justify that the Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA must be grounded in workload awareness and a consistent, criteria-based ranking process that give all employees a fair chance of being recommended, rather than a separate fairness or diversity scoring feature, in order to lessen the problem of repeated and unbalanced employee assignment.

Lastly, switching to automated system can improves employee's morale and productivity. Bai et al. (2022) found that work is being assigned by a fair algorithm make the employees often feels more satisfied because the process is transparent and unbiased, and with this the sixth need of the study is identified, which is the need for an automated system that improves employee morale through transparent and fair assignment process. Building on this, the seventh need is identified through Lima et al. (2024), who argue that automated and fair resource management is no longer optional but a strategic necessity for companies that want to retain skilled workers and maintain productivity, since the manual process of tracking who is actually free is making it hard, and this type of technology directly address that problem by removing the subjectivity from the assignment process. This is further supported by Banerjee et al. (2023), who point out that a fair recommender system must consider not just the company's needs but also the individual employee's stake in the assignment process,

because a system that only optimizes for project completion without considering employee availability and satisfaction will eventually lead to burnout and disengagement, which gives the eighth need of the study, the need for a recommender system that balances company output with employee wellbeing.

Transitioning from this individual perspective to a broader organizational view, Zhou et al. (2024) support the ninth need of the study, which is the need for a sustainability aware recommendation approach that considers the long term impact of resource allocation decisions on both the company and its employees, instead the manager are relying on the gut feeling or memory, a data driven system approach that actual looks at availability and long term sustainability of the workforce. And this strategic shift are already in the local landscape, a study by Cacatian et al. (2025) AI integration in Metro Manila's HR recruitment shows that automated innovation is being a necessity to remain competitive, and with this the tenth need of the study is identified, which is the need for WEA to align itself with the regional trend of adopting AI driven human resource tools to remain competitive in the Philippine industrial market. With these studies justify that adopting recommendations engine, WEA place itself with the regional trends, that moves toward to modernized and efficient resource allocation framework.

The remaining studies in this topic consistently shows, the literature consistently shows that the shift from manual to automated resource allocation is not just a technical upgrade but a strategic and cultural necessity. Bai et al. (2022) and Lima et al. (2024) establish that employee morale and organizational productivity are directly affected by the fairness and transparency of the

assignment process, while Banerjee et al. (2023) and Zhou et al. (2024) further confirm that a truly effective recommender system must also give attention to the short term project needs and the long term wellbeing of the workforce, even though the current version of the system does not yet include a dedicated long-term sustainability model. And the local evidence from Cacatian et al. (2025) shows that this shift is already happening in the Philippine context, making it even more urgent for WEA to adopt this technology. With all of these five needs justify that the Resource Recommendation System is not only technically necessary but also strategically and culturally aligned with the direction that modern organizations are taking in managing their human resources more transparently and consistently, with fairness treated as an outcome of applying the same skill, workload and performance criteria to every employee rather than a separate scoring feature. In the context of WEA, this is especially relevant because the company mentioned that the same people always end up being assigned to multiple projects at the same time because the leads do not have a system to track who is already loaded.

Skill Matching and Employee-Project Alignment

Matching the right employee to the right project is one of the important part in making the resource management work properly. The process of identifying who is qualified for a specific task is not just about checking if the employee has the skill. It is also about making sure that the skill matches the

exact requirement of the project. According to Mueller and Schaaff (2025), the AI based system for matching employees to project roles must not only consider the current skill level of the employee, it must also consider the interest of the employee, because interest are an important motivator for workplace learning. With this, the first need of the study is identified, the need for an automated employee project matching system that goes beyond just checking a single skill match and also considers additional employee attributes such as availability and performance history, in order to produce better and more accurate assignments. Building on this, the second need is identified through Tang et al. (2025), who points out that person job recommendation systems are widely used in human resource management, but many of them are considered as black box systems because their internal decision making processes remain impenetrable to the users. This is very relevant to WEA because the team leads and the resource manager will only trust and use the system if they can clearly understand the basis of every recommendation, making transparency and explainability a necessary requirement for the system.

The third need of the study is identified through the findings of Boukhari et al. (2024), who discuss that the skill matching at scale requires a system that can handle different types of candidate profiles and project descriptions in an efficient way. This is relevant to WEA because even if the company currently have a small team, the system must still be designed in a way that it can handle more employees and more projects as the company grows. Kashif and Kumar (2024) found that using NLP and machine learning to extract resume content and

analyze it can provide actionable feedback and improve the accuracy of matching candidates to the right roles, which support the fourth need of the study, the need for an NLP based resume extraction system that can automatically process employee documents and produce structured and searchable skill profiles for matching. Lastly, the fifth need is identified through Kishore and Sreerala (2025), who found that AI powered job matching systems that uses NLP and large language models can minimize the manual effort of reviewing resumes, and can also improve the accuracy of matching candidates to the right positions. This shows that technical skills, experience and job keywords can be extracted from resumes, allowing more accurate candidate profiling that the system can use to compare against the project requirements.

What the literature reveals about skill matching and employee project alignment is that the core problem of WEA in matching employees to projects is not just a matter of inefficiency, it is also a matter of accuracy and transparency. Mueller and Schaaff (2025) establish that a good matching system must go beyond just checking skills and must also consider the full profile of the employee. Tang et al. (2025) strengthen this by arguing that transparency in the recommendation process is a critical requirement for gaining the trust of the users. Boukhari et al. (2024) further confirm that the system must be scalable and capable of handling different types of documents and profiles, while Kashif and Kumar (2024) and Kishore and Sreerala (2025) both prove that NLP and machine learning based extraction is a reliable and accurate alternative to the manual review that WEA is currently doing. With all of these five needs, it justify

that the Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA must be grounded in automated matching, transparent recommendations and automated skill extraction, in order to truly solve the problem of slow and inaccurate employee project assignment.

The way skills are being represented and ranked in the system also affect how the final recommendation will turn out, especially when multiple employees qualify for the same project. Naveed et al. (2024) found that using Natural Language Processing with similarity measures like Jaccard Similarity can extract the competencies of the employees from their resume and match them to predefined skill sets associated with the project requirements. This is very relevant to WEA because the team leads mentioned that when two people can both do the task, they consider who has more relevant skills and more experience before making the final assignment. This gives the sixth need of the study, the need for a skill overlap and similarity measurement mechanism that can rank employees based on how closely their extracted skills match the specific requirements of the project. Building on this, the seventh need is identified through Decorte et al. (2024), who discuss that evaluating the relatedness between skills using self supervised learning can improve how the system understand the connection between different technical skills, even when they are expressed in different ways, which is a common situation in WEA where the employees submit their documents in different layouts and file types.

The eighth need of the study is identified through the findings of Clavié and Soulié (2023), who found that large language models can be used as zero

shot skill matchers, ones that can identify and match skills from job descriptions and candidate profiles even without requiring extensive training data. Because of this, the system can handle the technical and specialized skills that are common in the electrical distribution industry of WEA, like lighting calculations, junction box installations, and safety certifications, even when these skills are expressed in non standard terms or abbreviations. Transitioning from this technical skill matching perspective, the ninth need is identified through Magron et al. (2024), who discuss that generating synthetic and structured job postings and skill profiles can enhance the accuracy of skill matching. This is relevant to WEA because the company currently do not have a structured way of storing and comparing the skills of their employees. Lastly, the tenth need is identified through Gavrilescu et al. (2025), who found that using keyword extraction and transversal skill classification from job advertisements can help the system identify and organize the specific technical skills and competencies required for different types of projects, making it easier for the resource manager to see at a glance who is qualified and who is not, without having to manually check every document.

Building on this, the eleventh need of the study is identified through the findings of R et al. (2026), who found that combining TF-IDF based feature extraction with a Logistic Regression classifier can produce an overall resume classification accuracy of 88%. This shows that the matching process must also include a validation step using classification techniques, to make sure the extracted skills are truly relevant before it is being used in the recommendation.

This is supported by Saatçı et al. (2024), who discuss that resume screening using Natural Language Processing still need a proper mechanism to categorize and confirm the skills that are extracted, especially when the resume contain different formats and terminologies. Frazzetto et al. (2025) further support this by showing that representing candidate profiles through structured embeddings and similarity measures can improve how accurately a system matches candidates to specific job requirements, proving that a structured representation approach is a reliable technique in comparing the extracted skills against the project requirements. Lastly, Varshith Reddy et al. (2025) found that a resume analyzer and job recommendation system that includes a validation and classification component can produce more reliable and accurate job matching results. This give the twelfth need of the study, the need for the system to include a TF-IDF based representation and Logistic Regression based validation step after the skill extraction, to confirm the accuracy of the matched skills before it is presented to the Resource Manager as a recommendation.

Taking a closer look at the remaining studies in this topic, the shift from manual to automated skill matching is not just a technical upgrade, it is also a practical necessity for organizations like WEA that handles technical and safety critical projects. Naveed et al. (2024) and Decorte et al. (2024) establish that similarity measurement and skill relatedness evaluation are essential components of an accurate matching system. Clavié and Soulié (2023) further confirm that even specialized and non standard technical skills can still be matched accurately using modern language models, without the need of

requiring large amounts of training data. Magron et al. (2024) and Gavrilesu et al. (2025), on the other hand, shows that structuring and classifying skills in a consistent way is the key in making the matching process reliable and scalable as the company grows. This is strengthen further by R et al. (2026), Saatçı et al. (2024), Frazzetto et al. (2025) and Varshith Reddy et al. (2025), who shows that using TF-IDF for representing skills, together with Logistic Regression for classification and validation, are proven and reliable techniques to confirm that the matched skills are accurate and relevant before it is being use in the recommendation. With all of these five needs, it justify that the Resource Recommendation System must be equipped with a robust and automated skill matching engine, one that can handle different document formats, recognize technical and specialized skills, and produce ranked and transparent recommendations that the resource manager can understand, verify and confidently act upon.

Optical Character Recognition (OCR) with NLP in Recommender Systems

Text recognition and extraction technology integration is one of the major step making recommendations in system are more efficient and automated. Combining Optical Character Recognition (OCR) with the personalized algorithms allows system to process information and user to make more informed decisions, according to Zhang et al. (2023), which is important in organizations like WEA, where employee qualifications like skills must be verified quickly and

accurately, and manual data entry is often time consuming and prone to error. With this the first need of the study is identified, which is the need for an automated document processing system that can convert different types of employee documents like scanned resumes into a readable and searchable text that the system can use for matching and recommendation. Building on this, Helli et al. (2023) found that combining OCR with NLP models to extract information from resumes after preprocessing can significantly improve the accuracy and efficiency of the information extraction process, showing that the automated document processing is reliable enough to replace the time consuming manual review of the employee internal resume that the team leads are currently doing in WEA.

The second need of the study is identified through the findings of Najam and Faizullah (2023), who shows that modern deep learning architectures can achieve the error rate as low as 2.03% in recognizing and extracting text from documents, which support the need to reduce delays in reviewing employee documents especially in WEA where the review of skills and qualifications must be done before every project assignment. Transitioning from this accuracy perspective, Thangam et al. (2023) further support the third need of the study, which is the need for a post processing mechanism that can clean and improve the raw text output of the OCR system so that the NLP component can still process it correctly even when the extracted text contains errors, noise or inconsistencies, which is a real situation in WEA because the employees submit their documents in different formats and quality levels. Malashin et al. (2024)

report how the adaptive system can adjust themselves based on the document's layout to improve the extraction, giving the fourth need of the study, the need for an OCR system that can apply preprocessing techniques such as grayscale conversion, contrast enhancement and deskewing to automatically adjust to different document layouts and formats. Lastly, Alsaif et al. (2022) found that advanced NLP techniques do not need perfect OCR text to successfully match jobs to candidates, and this is supported by Gan et al. (2024), whose LLM based resume screening framework reached an F1 score of 87.73% in classifying resume sentences and turning raw resume text into structured, usable candidate information, which gives the fifth need of the study, the need for an NLP based text processing system that can convert the raw and unstructured text extracted by the OCR into meaningful and structured information.

The studies gathered on OCR and NLP document processing collectively shows, the literature consistently shows that the core problem of WEA in processing employee documents is not just a matter of speed but also of accuracy and reliability. Zhang et al. (2023) and Helli et al. (2023) establish that combining OCR with NLP is a reliable and efficient approach for automatically converting employee documents into structured and searchable information, while Najam and Faizullah (2023) strengthen this by proving that modern OCR systems can achieve very low error rates that are more than acceptable for a professional environment like WEA. Thangam et al. (2023) and Malashin et al. (2024) further confirm that even when the document quality is inconsistent, the system can still produce accurate results through post processing and

preprocessing based extraction techniques, and Alsaif et al. (2022) and Gan et al. (2024) both prove that the NLP component can still match qualifications accurately even when the extracted text is not perfectly clean. However, a difference can be noted between Malashin et al. (2024) who focus on layout adaptation and Thangam et al. (2023) who focus on post processing correction, showing that the two approaches address the same problem from different angles and both must be considered in building a robust document processing pipeline for WEA. With all of these five needs justify that the Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA must be grounded in a robust and automated OCR and NLP pipeline that can handle different document types and qualities to truly solve the problem of slow and inconsistent manual document review.

Lastly, WEA currently has a small team and the leads already know the skills of each person informally, the OCR and NLP features become more important as the company grows or when new employees are onboarded. Janani et al. (2025) found combining OCR with machine learning algorithms can produce more accurate profiles, which gives the sixth need of the study, the need for a scalable document processing system that combines OCR with a trained classification algorithm so the system can produce more accurate and complete employee profiles as the number of employees and documents increases over time. Building on this, Biol and Abalorio (2024) used a fine-tuned Named Entity Recognition or NER model to parse unstructured resume data into searchable profiles, identifying the technical skills, education and qualification details from

the internal resumes even when the documents are not in a standard format, which gives the seventh need of the study, the need for a Named Entity Recognition component that can specifically identify and extract the technical skills and qualification details from the unstructured text. Sharma et al. (2023) found that hybrid NLP based methods can achieve up to 92.7% accuracy in matching skills from resumes to job requirements, which is more than enough to replace the manual review process that the team leads are currently doing, giving the eighth need of the study.

Transitioning from this technical accuracy perspective, the ninth need of the study is related to the overall impact of combining all of these technologies together in one system. Zhang et al. (2023) discuss that the integration of OCR with personalized recommendation algorithms allows the system not only to extract information but also to use that information to make more informed and accurate decisions, giving the ninth need, the need for an integrated OCR and recommendation system that can extract, process and directly use the information in the recommendation engine. Lastly, Helli et al. (2023) also found that classifying text groups from OCR processed resumes using NLP models can produce structured and organized employee profiles that the system can store and use for future matching, giving the tenth need of the study, the need for a structured employee profile storage system so the team leads does not need to re upload and re process the documents every time a new project assignment is needed.

The literature consistently shows that the shift from manual to automated document processing is not just a technical upgrade but a strategic investment in the long term operational capacity of WEA. Janani et al. (2025) and Biol and Abalorio (2024) establish that combining OCR with a trained classification algorithm and Named Entity Recognition is essential for producing accurate and complete employee profiles that the system can use for matching and recommendation, while Sharma et al. (2023) further confirm that hybrid NLP methods can achieve accuracy rates that are more than sufficient to replace the manual review process that the team leads are currently doing. But a difference can be noted between the findings of Janani et al. (2025) who focus on a general classification approach for profile building and Biol and Abalorio (2024) who focus on Named Entity Recognition for skill extraction, which shows that there is no single best approach and the system must combine both methods to achieve the most accurate and complete employee profiles. Zhang et al. (2023) and Helli et al. (2023) both prove that integrating OCR with recommendation algorithms and structured profile storage can transform the document processing pipeline into a fully automated and scalable system that WEA can rely on even as the company grows and onboards new employees in the future. With all of these five needs justify that the Resource Recommendation System must be equipped with a complete and automated OCR and NLP pipeline that can extract, process, classify and store the information from different types of employee documents, so that the system can produce faster, more accurate and more reliable employee

assignment recommendations that the team leads can trust and confidently act upon.

Rapid Application Development (RAD) for Efficient System Deployment

Rapid Application Development or RAD is now well known and recognized as an effective software development methodology because it gives fast prototyping making the work more easily and efficient to the users. According to Novita et al. (2023) and Meyliana et al. (2024), it shows that Rapid Application Development helps produce successful systems while reducing the development risks, particularly when users are involved early in the design and testing stages, and with this the first need of the study is identified, which is the need for a development methodology that can involve the users of WEA early in the design and testing stages so that the system can be adjusted based on the actual needs of the team leads before the final version is completed. Building on this, Parlika et al. (2023) further support by showing that RAD is consistently identified as the most efficient model in software development because it has the shortest processing time compared to other methodologies, giving the second need of the study, the need for a development methodology that can produce a working system in the shortest possible time so that the team leads of WEA can start using the workload tracking and recommendation features as soon as possible. The third need of the study is identified through the findings of Brill (2025), who explains that RAD is an iterative approach that focuses on quick functional

prototypes, regular user feedback and flexibility in changing requirements, making sure that the final version of the system truly reflects the actual workflow and assignment process that WEA is currently using.

Building on this, Riadi et al. (2024) found that the RAD method has a significant positive impact on both the development cycle and user satisfaction, particularly in web based systems where the users need to interact with the system regularly, giving the fourth need of the study, the need for a development methodology that can ensure a high level of user satisfaction so that the team leads of WEA will actually use the system consistently after it is deployed instead of going back to the manual process. Lastly, the fifth need is identified through Shaker et al. (2023), who shows that the adaptability of RAD allows it to be applied not only in information systems but also across various domains such as mental health services and healthcare application development too because of the flexibility of Rapid Application Development, supporting the development of the Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA, as the project requires rapid development and the ability to adapt features based on both company requirements and user needs.

Taking these first five findings together, what stands out the most is how RAD keeps proving itself as a methodology that fit both the pace and the practical limitation of a project like this one. The early user involvement that Novita et al. (2023) and Meyliana et al. (2024) point out, combine with the efficiency that Parlika et al. (2023) found, already give a strong basis for why RAD suit the needs of WEA. Brill (2025) and Riadi et al. (2024) add another layer

to this by showing that the iterative and prototype driven side of RAD does not only speed up development but also keep the users satisfied and engage with the process. Riadi et al. (2024) lean more toward user satisfaction while Parluka et al. (2023) is more focus on how efficient the development timeline can be, and this difference actually work well together, since it prove that RAD is not a one dimensional solution but something that address speed and user experience at the same time. Taking all of these five needs into account, adopting RAD is not just the practical route to take but also the one that is strategically align with what WEA actually requires from its development process.

On the other hand, not all findings paint RAD in a fully positive light. Carnovalini (2025) point out that RAD sometimes focus too much on speed, which can put the systems scalability and maintainability at risk in the long run if there is no proper planning behind it, and this identify the sixth need of the study, which is the need for a careful and well documented development plan that complements the rapid pace of RAD to ensure that the system remains reliable and scalable even after deployment. Wang et al. (2025) build on this concern by showing that functional prototypes made through RAD can help catch and fix problems early, but this only work if the testing process itself is done thoroughly, giving the seventh need of the study, the need for the system to be tested carefully at every iteration to ensure that the OCR, NLP and recommendation components are all working correctly before they are integrated into the final system, which is very relevant to WEA because any error in the recommendation output could directly affect the operations of the company.

Imran and Bhutto (2023) contribute the eighth need by finding that RAD based web application development work particularly well for small and medium organizations since the methodology is designed around small teams and limited resources, which describe well the situation of a small research team developing a system for a small company like WEA. Stankovic (2024) then identify the ninth need through his discussion on how RAD in modern software engineering remain relevant and effective, especially for projects that need fast delivery paired with continuous user involvement. The tenth and final need come back to Carnovalini (2025), who warns that the risk of prioritizing speed over quality in RAD has to be manage properly through careful planning and documentation.

Looking at this second set of studies, its clear that the success of RAD depends not just on how fast a system get built but also on how well the planning, testing and user involvement is being handled all throughout the process. Carnovalini (2025) already warns that without a proper planning, the speed that RAD is known for could turn into a liability instead of an advantage. Wang et al. (2025) and Imran and Bhutto (2023) both emphasize as well that a thorough testing of the prototype, together with a development approach that fit the size and capacity of the organization, are factors that really determine if a RAD based system will perform good in a real world environment such as WEA. Stankovic (2024) meanwhile look at RAD from a different angle, focusing more on how relevant RAD still is in todays modern software engineering, compare to Carnovalini (2025) who is more focus on the danger of prioritizing speed over quality. This only proves that RAD must be apply carefully especially in a system

that will be use for critical project assignment decisions like in WEA. With all of these ten needs identified in this topic, it justify that the Resource Recommendation System must be build using a well planned and carefully documented RAD process, one that balance speed with quality, and rapid prototyping with a thorough testing, so the final system will not only be built quickly but also correctly and reliably for the long term use of WEA.

Evaluating System Effectiveness

To see if the transition from manual to automated resource management is successful, the system must be measure against the international standards. The ISO/IEC 25010 standard will serve as the standard framework in evaluating that will allows researchers to analyze both the technical functionality and user satisfaction (Nugraha and Rachman, 2024), and by applying this standard, the study can move beyond subjective opinions and provide the assessment on how the system performs in real word corporate environments. With this the first need of the study is identified, which is the need for a standard evaluation framework that can measure how well the system works compared to the manual process that WEA is currently doing. Building on this, Rojas et al. (2025) shows that ISO/IEC 25010 has been consistently used across different domains and industries which proves that the standard are flexible and reliable enough to be apply in different organizational context including the resource management context of WEA, giving the second need of the study, the need for an evaluation

framework that are not just technically rigorous but also widely accepted across different fields so that the findings of the evaluation can be considered credible.

There's literature that demonstrates high performance in the quality areas is achievable for institutional systems. Kasri et al. (2024) found that functional suitability and reliability score as high as 98% and specialized information system as 100%, giving the third need of the study, the need for the system to achieve a high level of functional suitability and reliability that are comparable to the benchmark scores in similar institutional systems. Similar to Nuzula and Rochimah (2023) that found human resource management tools can achieve 90% in appropriate and completeness in performance, which suggest different quality characteristics are interconnected, giving the fourth need of the study, the need for the system to perform consistently well across all the quality characteristics because a weakness in one area can affect the overall performance of the system. Lastly, Barletta et al. (2023) highlight that evaluating factors such as safety, response time and context help demonstrate the quality from the end user's point of view, giving the fifth need of the study, the need for the system to be evaluate not just from a technical perspective but also from the perspective of the actual users of WEA who will be using it on a daily basis.

The further studies in this topic strengthens, the literature consistently shows that the ISO/IEC 25010 standard is the most appropriate framework for evaluating the quality of the Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA. Nugraha and Rachman (2024) establish that ISO/IEC 25010 provides a comprehensive framework for

measuring both technical functionality and user satisfaction, while Rojas et al. (2025) strengthen this by showing that the standard are widely used and validated across different domains. Kasri et al. (2024) and Nuzula and Rochimah (2023) further confirm that high performance scores are achievable for institutional and human resource management systems that provides a meaningful benchmark for the system to aim for. But a difference can be noted between Kasri et al. (2024) who focus on the functional suitability and reliability and Barletta et al. (2023) who focus on the quality from the end user's point of view, which shows that evaluating the system from both the technical and the user perspective is necessary to get the complete picture of the system quality. With all of this studies justify that the evaluation of the Resource Recommendation System must be grounded in ISO/IEC 25010 and must cover both the technical performance and the user experience to see if the system is really a better alternative than the manual process that WEA is currently struggling with.

The technical reliability are only one part of the equation, what truly defines the system's impact is the quality of use from the human perspective. Gotz (2023) discuss that system with smaller datasets, a simpler recommendation can be highly effective without requiring decades of data history, giving the sixth need of the study, the need for the system to still produce reliable recommendations even if WEA do not have a long record of past project assignments since the company only relies on Excel files for tracking. Building on this, De Campos et al. (2024) and Alberto et al. (2025) both found that a system

that gives short, clear and explainable suggestions is the key for gaining the confidence of the resource manager, giving the seventh need, the need for the system to give clear explanation for every recommendation it makes so the team leads of WEA can understand why a certain employee is being suggested and can act on it without hesitation.

Lastly, the eighth and ninth needs are identified through the findings of Ilyas and Ahmad (2023) and Susanto and Andriana (2023), who both show that usability and functional suitability are the most critical quality characteristics especially for systems that are being used by non technical users, and that measuring the quality using ISO/IEC 25010 can help identify specific areas where the system still needs improvement before it is officially deployed. This is very relevant to WEA because the team leads who will be using the system are not IT professionals, and the system must be easy to navigate and understand for them to actually use it in their daily assignment process without needing technical support. And the tenth need is that the system must follow the ISO/IEC/IEEE 26514:2022 standard that ensures the user manuals and troubleshooting guides are clear and user friendly.

The studies reviewed on system evaluation consistently show, the literature consistently shows that the quality of use from the human perspective is just as important as the technical performance of the system. Gotz (2023) and De Campos et al. (2024) both show that even with small and limited data, the system can still produce useful and trustworthy recommendations if the suggestions are clear and directly connected to what the user actually needs,

while Alberto et al. (2025) further support this by showing that the clarity and relevance of the recommendations are what makes the users actually accept and use the system in their daily work. But a difference can be noted between Ilyas and Ahmad (2023) who focus on usability and functional suitability as the most important quality characteristics and Susanto and Andriana (2023) who focus more on using the evaluation results to find and fix the weak areas of the system before deployment, which shows that the ISO/IEC 25010 evaluation is not just a one time judgment but also a tool for making the system better. With all of this studies justify that the Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA must be evaluate in a way that covers both the technical quality and the practical usability of the system so that the team leads of WEA can truly rely on it every day to make faster, fairer and more accurate project assignment decisions than the manual process they are currently doing.

Performance-Based Evaluation and Portfolio Management in Employee Resource Allocation

Aside from skills and availability, an employee's actual track record of performance is also an important factor that a resource recommendation system must consider before assigning a project. Jemal et al. (2024) found that a recommender system that used the employee's past competency scores and profile data together with NLP techniques can predict future performance with

more than 80% precision, showing that historical performance record is a reliable basis in recommending who should be assign to a new project. With this the first need of the study is identified, which is the need for the system to use the employee's historical performance and client feedback record as one of the basis in generating the recommendation, and not just relying on the skills and availability alone. Building on this, Youssri et al. (2024) discuss that machine learning based performance evaluation that is integrated directly to an enterprise resource planning system can help managers identify the qualified employees faster, since the traditional performance review process is slow and cannot give a real time feedback of the employee's actual performance, this support the second need of the study, the need for the system to continuously update and analyze the performance data of the employee in real time. The third need of the study is identified through the findings of Yang (2025), who developed a framework that combine fuzzy logic and neural network to predict the performance trend of the employee and use it to determine how the resource should be allocated, achieving a high accuracy of prediction and proving that a data driven approach is more accurate compare to relying on the manager's own judgement, giving the third need, the need for the system to also analyze the pattern of employee's performance overtime, not just the current record.

Transitioning from the performance perspective to the portfolio perspective, Poncio (2024) reviewed different techniques use in matching an applicant's profile, that consist their collected experience, skills and past work, to available job opportunity, and found that a hybrid approach combining both the

primary information given by the user and the secondary information mine from other sources produce a more personalized and relevant result. This identifies the fourth need of the study, the need for the system to build and maintain a portfolio record for every employee that consolidates their skills, completed projects and client feedback in one place. Lastly, the fifth need is supported by Hanif et al. (2025), who found a significant positive relationship between the implementation of a competency based performance assessment and the employee's actual performance and productivity, with a result of 68% variance explained by the assessment system, giving the fifth need, the need for the system to adopt a competency based performance assessment approach that is consistent and transparent, so it can be trusted by both the resource manager and the employees of WEA as basis for a fair project assignment.

The studies gathered in this topic consistently show that performance history and portfolio record are equally important as skills and availability in producing a fair and accurate recommendation. Jemal et al. (2024) and Yang (2025) establish that a data driven and predictive approach to performance is more accurate than manual evaluation, while Youssri et al. (2024) strengthen this by showing that real time performance evaluation supports a faster decision making. But a difference can be noted between Yang (2025), who focuses on predicting performance trend, and Poncio (2024), who focuses on consolidating the employee's portfolio for matching, which shows that performance tracking and portfolio management are related but different components. With all of these five needs justified, that the system for WEA must include a performance based

evaluation and portfolio management feature that track the historical performance, client feedback and completed project record of every employee.

Significance of the Study

For the organization (WEA), the system will help manage the technical workforce more effectively by reducing the time spent on manual checking of internal resume. This improvement minimizes the human error, enhance the safety compliance and support the company's goal of maintaining the high quality of project delivery. It also help the company connect their internal workforce with the external hiring process, so that when the current employees are not enough to cover a project, the company can immediately fill the gap with the applicants that are already screen by the system.

For the resource and project manager, the system made daily operation more manageable. They will be able to quickly identify the qualified employees who met the strict technical requirements of each project, that prevent scheduling conflicts and operational delays. This allow managers to make faster, more accurate decisions and especially when the organization grew and handle multiple number of complex projects.

For employees, the system will provide transparency and fair work environment. Their skills and qualifications will recognize accurately through the system ability to analyze different documents formats that ensure task

assignments is based on the actual availability. The focus is to ensure that employees are only assign to roles they are qualified for, increasing their professional confidence and safety in the job.

For the Human Resources personnel, the system made the applicant screening process more organize since the applicant's resume and their manually input details are shown together, this allow the Human Resources personnel to easily check if the input details match the actual resume before deciding who to filter and recommend. This allow the Human Resources personnel to spend more time on the interview and the final hiring decision instead of manually going back and forth between the resume file and the application form.

For the applicants, the system will give them a fair chance in applying for a position in WEA since their recommendation is based only on the skills and qualifications that they submitted, not on personal connections or how the application form was reviewed manually. This ensure that every applicant is given equal opportunity to be consider for employment based on their actual qualification.

For the researchers, the study will serve as opportunity to enhance their technical and competencies. Developing a system that integrates complex AI tools like OCR and NLP, the researchers will able to apply theoretical programming knowledge to solve the real world industrial problems. With this project, it allows them to gain deeper understanding of the human resource

management and safety standards of oil and gas industry that will prepare them for professional careers in software development and project management.

Lastly, the future researcher, this study will serve as useful reference for someone who are interested in exploring AI driven human resource management. The integration of the OCR and NLP for specialized industries will provide foundation for developing similar intelligent system in other fields. Future developments can build by remodel the system to include features like automated attendance, payroll integration or real time tracking of license expirations.

Statement of the Problem

This study aims to develop an automated skill extraction and workload-based resource recommendation system for WEA to improve workforce allocation, task matching, and operational efficiency.

1. What system can be developed for WEA to automate skill extraction and support intelligent resource allocation?
2. How to extract and classify employee skills from internal resume documents via OCR and NLP?
3. How to analyze workload distribution across employees to identify underutilization and overloading?

4. How to generate accurate and fair resource recommendations based on skill matching, availability, and workload conditions?
5. How to continuously learn and adapt from historical assignments and performance outcomes to improve recommendation accuracy?
6. How can data analytics be integrated to provide insights into workforce capacity, skill gaps, and strategic planning?
7. What is the level of acceptability of the Skill-Extraction and Workload-Based Resource Recommendation System based on the ISO/IEC 25010 software quality model in terms of:
 - a. Functional Suitability
 - b. Reliability
 - c. Usability
 - d. Performance Efficiency
 - e. Maintainability
 - f. Portability
 - g. Security

Scope and Limitation of the Study

The focus of this study is the development and evaluation of the Resource Recommendation System that is design exclusively for WEA's workforce allocation process, a small electrical distribution company with fourteen to fifteen employees. The system uses Optical Character Recognition or OCR, a technology that convert the employee internal CV into machine-readable text, and Natural Language Processing or NLP, a field that enable the computer to understand and process human language, that will extract and normalize the employee skills from the extracted text. This OCR and NLP extraction process only apply to the internal CV of the employee, since the internal CV follow the standardized format that WEA is recommend for its employees. The system also tracks the employee workload and determines the employee availability in real time, this information together with the extracted skills, availability and the historical performance of the employee are being used by the system to recommend and rank the employees that are most suitable for the project assignment. The historical performance is base on the feedback given by the client after every project completion, and this is combine with the skill matching, workload and availability to compute the final recommendation score.

Aside from the internal workforce allocation, the system also cover the external hiring process of WEA through the Human Resources function. This include the posting of job openings, the filtering of applicants who submit their application form and resume, and the reviewing of the applicants before the Human Resources personnel make the final hiring decision. The applicant's

resume is only upload as a supporting document, and the applicant is required to manually input their own information such as skills and other relevant details through the application form, this information is what the system uses to filter and recommend the applicant to Human Resources. This expansion allows the system to supplements the internal talent pool that the Resource Manager can draw from, since the recommended and hired applicants can eventually be considered for future project assignment once they are already part of the company's employee record.

The system only accepts the standardized internal CV format for the OCR and NLP based extraction, other resume format that are not following the standard layout are not supported by the system for automated processing. The study is develop using the Rapid Application Development or RAD methodology to allow iterative design, prototyping and user feedback during the development process. The system performance is also evaluated using the ISO/IEC 25010 software quality standard, that focus on the functional suitability, reliability, usability, performance efficiency, maintainability, portability, and security of the system.

This study has several limitations. The accuracy of the OCR component is depends on the document quality, because of this handwritten documents and low resolution scans are outside the capabilities of the system to process, this means that the employee that will submit this type of internal CV may not be able to use the system properly. The NLP component is optimized for English language documents and may have difficulty interpreting highly specialized

regional terminology and non standard abbreviations that are commonly used in the electrical distribution industry, this could affect on how accurate the system extract and interpret certain technical skills. The system only supports PNG, JPEG and PDF file formats for the internal CV, other file types like Word documents are not being process by the system, so the employee needs to convert their file first before it can be upload.

Since the applicant's information is manually input rather than extracted through OCR and NLP, the system does not verify if the skills and details that the applicant provide are accurate or true, the system only rely on the honesty of the applicant when they fill up the application form, the resume they upload only serve as a reference document and does not automatically populate or validate the manually entered fields, although the Human Resources personnel can still manually verify the applicant's resume against the manually entered information before the final hiring decision is made. The historical performance rating that is being used by the recommendation engine is only limited to the projects that are completed and rated within the system, this means that any performance or feedback the employee received before the system was implemented will not reflect on their performance score, and a new employee with no rating history yet will only receive a neutral score until enough client feedback is accumulated. The client feedback and performance rating is also dependent on the subjectivity of the client who provide it, the system does not validate or verify if the rating given by the client is fair or accurate, it only stores and use whatever rating is submitted. For the Human Resources and applicant module, the system only

handle the posting of job opening, filtering and recommendation of applicant, the actual interview scheduling, background checking and final employment contract processing are not cover by the system and are still done manually by the Human Resources personnel outside the system.

To comply with the Data Privacy Act of 2012, the study uses synthetic data during the testing phase instead of the actual employee internal CV, applicant information and client feedback of WEA, this is done to protect the privacy and confidentiality of the personnel and client records of the company.

The system is also designed exclusively for WEA and requires an active internet connection since it is a web based platform, without an internet connection the system will not able to process or extract any data. The system also does not include other business function such as the payroll, attendance and financial accounting modules, the system is only limited to the resource and workforce allocation process, including its extension to the external hiring and performance feedback loop. Lastly, the system does not verify the accuracy or the authenticity of the skills that are listed in the internal CV, the responsibility of validating the skills before uploading the internal CV into the system lies with the user who is uploading the document, the system only extracts and process the information that are already contained in the uploaded internal CV. The accuracy of the classifier used in the skill extraction also depends on the amount of human-verified feedback that is accumulated over time.

CHAPTER 2

METHODOLOGY

Research Design

The study uses Descriptive-Developmental research design because it focuses on both building technological solution and business problem. According to Meyliana et al. (2024), a developmental research design is a systematic process that focuses on designing, developing, and evaluating products to ensure they are effective and consistent. In the other hand, the descriptive research part of this design allows researchers to study and explain the current manual inefficiencies at the company, and the developmental part focus in the actual creation and testing of the Resource Recommendation System. The system is built using Rapid Application Development methodology, Meyliana et al. (2024) explained it is an iterative process that focuses on fast prototyping and user involvement to speed up development.

This design is most suitable choice because it provide clear path to identify the company's specific needs first and then develop the OCR and NLP tools that required to solve the real world delays and errors identified in the study.

Architectural Framework

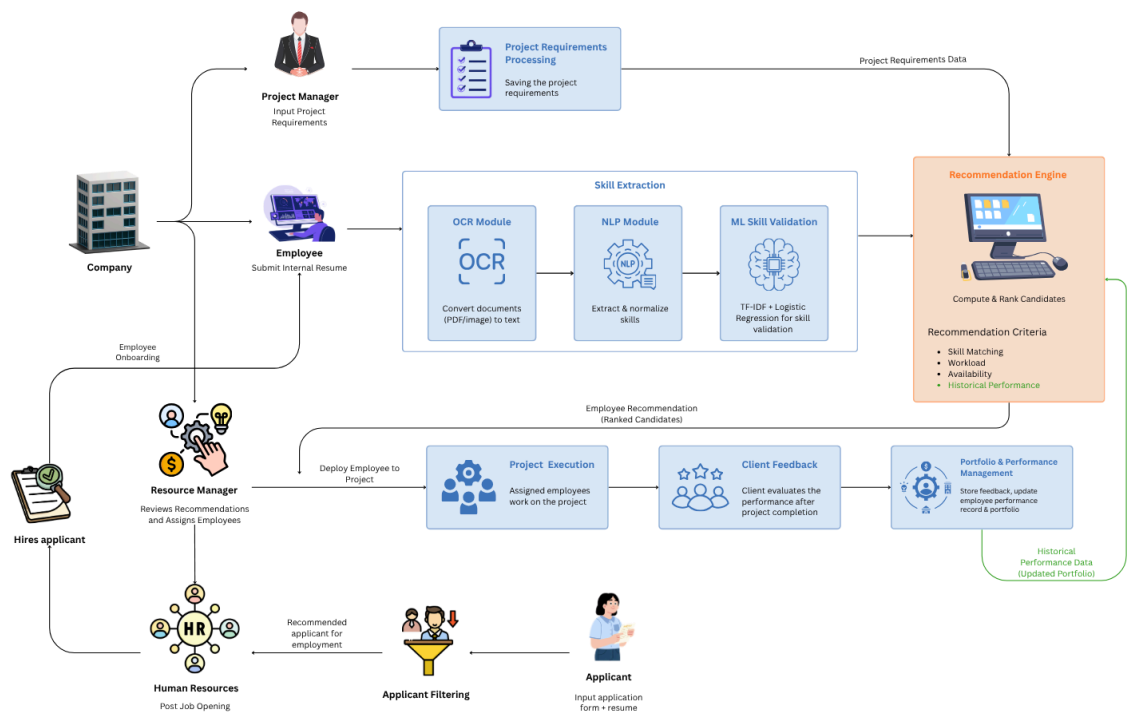


Figure 1. Architectural Framework

Figure 1 shows the architectural framework of the Resource Recommendation System. It explains the flow from the Company's Project Manager inputting project requirements, which are saved through Project Requirements Processing, and Employees submitting their internal resumes, which are processed through the Skill Extraction module comprising an OCR Module that converts documents into text, an NLP Module that extracts and normalizes candidate skills, and an ML Skill Validation step (TF-IDF + Logistic Regression) that confirms each extracted phrase represents a genuine skill. Both the structured project requirements and the validated employee skill data are

then passed to the Recommendation Engine, which computes and ranks candidates based on a defined set of recommendation criteria skill matching, workload, availability, and historical performance.

The Recommendation Engine returns a ranked list of employee recommendations to the Resource Manager, who reviews the suggestions and assigns the most suitable employees, deploying them to the project. Once Project Execution is underway and completed, the client evaluates the assigned employees' performance through the Client Feedback stage. This feedback is then passed to Portfolio & Performance Management, which stores the evaluation and updates each employee's performance record and portfolio. This updated historical performance data is fed back into the Recommendation Engine, allowing future recommendations to be informed by employees' actual track records rather than static profile data alone closing the loop between recommendation, deployment, and outcome.

The Company's Human Resources function manages external hiring: posting job openings, filtering applicants who submit their application and resume, and reviewing recommended applicants to make a final hiring decision, which supplements the internal talent pool the Resource Manager can draw from.

Level 0 DFD/Schematic Diagram

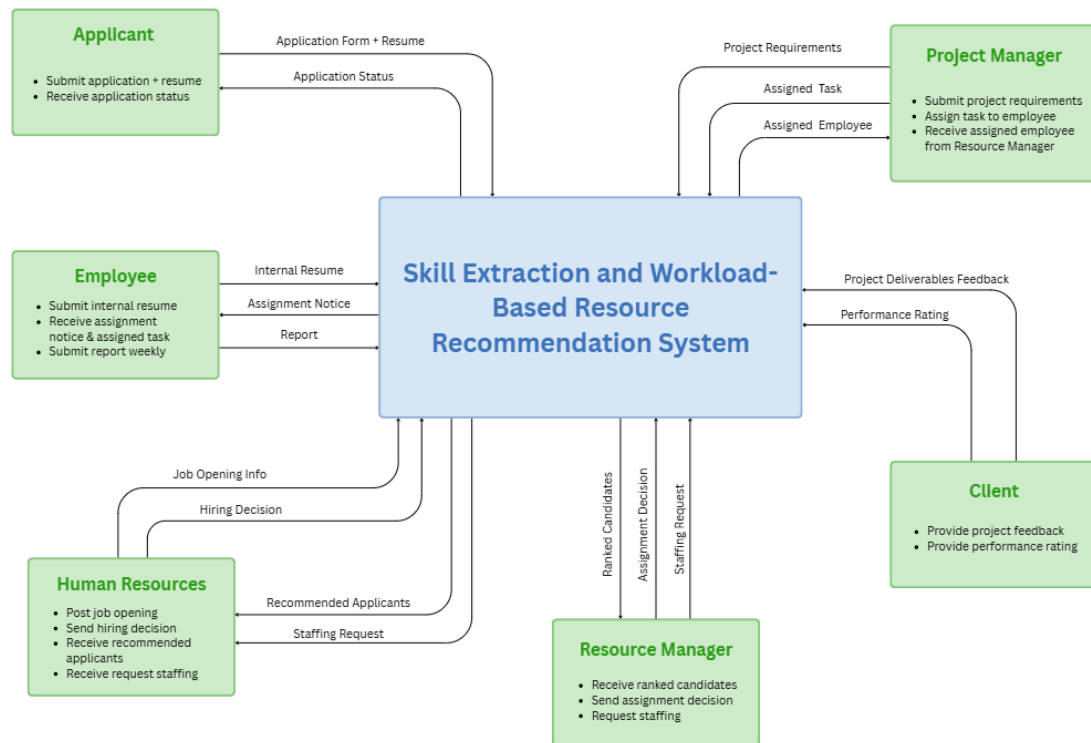


Figure 2. Data Flow Diagram (DFD)

Figure 2 shows the Data Flow Diagram (DFD) for the Resource Recommendation System, illustrating the data exchanged between the central system and its six external entities: Applicant, Employee, Project Manager, Client, Human Resources, and Resource Manager.

The Applicant submits an application form and resume into the system and receives back an application status. The Employee submits an internal resume and, in return, receives an assignment notice and assigned task from project manager, and separately submits a weekly report to the system. The

Project Manager submits project requirements and assigns tasks to employees, then receives back the assigned employee once the Resource Manager completes the assignment. The Human Resources entity posts job opening information and sends hiring decisions to the system, while receiving recommended applicants and staffing requests. The Resource Manager receives ranked candidates from the system, and sends back an assignment decision along with a staffing request when additional hires are needed. Finally, the Client provides project deliverables feedback and a performance rating, which the system uses to update employee records.

Algorithm Discussion

The Automated Skill Extraction and Workload-Based Resource Recommendation System follows the flow shown in the system's context diagram: the Employee submits an internal resume, the Project Manager submits the project requirements, and both are processed through Skill Extraction, which covers OCR and NLP. The extracted skills are matched against the project requirements inside the Recommendation Engine, and the Recommendation Engine sends the ranked candidates to the Resource Manager, who makes the final assignment decision. Once the employee is deployed, the Client's Performance Rating and Project Deliverables Feedback are stored and fed back into the system so that future recommendations are improved. This section

explains each stage of the algorithm step by step, and a short example is given after every step so the process is easy to follow.

Skill Extraction

Skill Extraction is the module responsible for converting an employee's uploaded document, whether a resume or a license, into a clean and structured list of skills. It runs in two steps: the OCR first turns the document image into readable text, then the NLP turns that raw text into validated and categorized skills.

Step 1: Optical Character Recognition (OCR)

When an employee submits an internal CV, the system first attempts to extract the text directly from the file using PDF parsing libraries, which works instantly for native, digitally-generated PDF documents. If this direct extraction fails or returns too little readable text which indicates that the document is a scanned image rather than a digital PDF, the system falls back to the OCR Module, which uses Tesseract OCR through Pytesseract to read the image and convert it into raw text. Before OCR is applied, each page is deskewed to correct any tilt, and the system generates two candidate versions of the image, a minimal version and an enhanced version. The system automatically selects whichever version produces higher OCR confidence for that specific page, so that clean pages are processed quickly while poor-quality scans still benefit from the extra enhancement. The output of this step is unstructured text that still

needs to be interpreted, since it is simply the words as they appear on the document, with no understanding yet of which parts are the actual skills.

Example: *An uploaded resume that reads “JOHN DELA CRUZ | Electrical Technician | SKILLS: AutoCAD (2D and 3D), PLC Programming with Siemens S7, Electrical Troubleshooting & Maintenance is converted by the OCR Module into that same text in a plain, machine-readable form, ready to be passed to the NLP Module.*

Step 2: Natural Language Processing (NLP)

Once the raw text is available, the NLP Module takes over to identify which parts of the text are the actual skills, clean them up, and organize them into categories. Instead of relying only on a fixed list of predefined skills, the system uses a dynamic pipeline made up of seven smaller steps, explained below, so that it can learn new skills, synonyms, and phrasing over time rather than being limited only to the skills.

Step 2.1: Tokenization

The raw text from the OCR is broken down into individual words and phrases using the spaCy NLP library. This gives the system a list of candidate tokens to examine, without deciding yet which of them are the meaningful skills.

Example: *The text “JOHN DELA CRUZ | Electrical Technician | SKILLS: AutoCAD (2D and 3D), PLC Programming with Siemens S7, Electrical Troubleshooting & Maintenance is tokenized into: “John Dela Cruz”,*

“Electrical Technician”, “AutoCAD”, “2D”, “3D”, “PLC Programming”, “Siemens S7”, “Electrical Troubleshooting”, “Maintenance”.

Step 2.2: Noun Phrase Extraction (Chunking)

Not every token is useful, since names, dates, and license numbers are not skills. This step uses dependency parsing to group the tokens into noun phrases, and keeps only the phrases that behave like skills, which are built from nouns, verbs, or adjectives that describe an ability or a tool, while filtering out names, dates, and other non-skill text.

Example: *From the tokens above, the system keeps “Electrical Technician”, “AutoCAD”, “PLC Programming”, “Siemens S7”, “Electrical Troubleshooting”, and “Maintenance” as noun phrases, while dropping “John Dela Cruz” (a person’s name) and “15 May 2023” (a date).*

Step 2.3: ML-Based Skill Classification

Each remaining phrase is passed through a trained classifier, specifically a TF-IDF feature representation with a Logistic Regression model, that returns a confidence score for how likely the phrase is a real skill. The system reads the score in tiers: a score of 0.75 and above is auto-approved as a skill right away; a score from 0.50 up to 0.75 is treated as a “likely skill” and is set aside for review instead of being auto-approved; and a score below 0.50 is rejected as not a skill.

Example: *“AutoCAD” scores 0.95, “PLC Programming” scores 0.92, “Siemens S7” scores 0.88, and “Electrical Troubleshooting” scores 0.90,*

all auto-approved since they clear the 0.75 mark. "Maintenance" scores 0.60, which lands in the "likely skill" tier, so instead of being auto-approved it is passed to the next step for a closer look.

Step 2.4: Employee Skill Confirmation

The phrases that land in the “likely skill” or “uncertain” tiers are not auto-approved, but they are not simply discarded either. These are surfaced to the Employee through the platform's skill-review screen, where each flagged phrase can be approved as a genuine skill or rejected as noise, before it is added into the structured skill profile. This keeps a person in the loop for exactly the phrases that the classifier is unsure about, while the confident majority, which are the phrases scoring 0.75 and above or the ones already sitting in the knowledge base, still pass through automatically without needing manual review.

***Example:** "Maintenance" sits at the "likely skill" tier with a confidence of 0.60, so it is shown on the Employee's review screen alongside the surrounding resume text for context. The Employee taps to confirm it is a genuine skill. The confirmed skill list becomes: AutoCAD, PLC Programming, Siemens S7, Electrical Troubleshooting, and Maintenance.*

Step 2.5: Model Retraining from Verified Feedback

Every approval or rejection that the Employee makes in the previous step is stored as a labeled example, either "Skill" or "Not Skill," and this human-verified set is the only ground truth that the classifier is trained on, since it

does not learn from unverified guesses. Once at least 20 new verified examples have accumulated since the last training run, the TF-IDF and Logistic Regression classifier is retrained on the full set, so its confidence scores keep improving the more employees confirm or correct its suggestions over time. Before that threshold is reached, or if a trained model is not yet available, the system relies on simpler rule-based checks instead of the classifier.

***Example:** After several employees confirm phrases such as "Maintenance" and "Cross-functional Collaboration" as valid skills, and reject phrases such as a person's name or an ID number as not skills, the number of newly verified feedback rows reaches 20. The classifier retrain on the full accumulated dataset, and the next time a similar phrase appears, its confidence score reflects what the employees have actually confirmed rather than the model's original, untrained estimate.*

Step 2.6: Deduplication and Aliasing

Different documents may describe the same skill using slightly different words. This step compares the phrases using spaCy's word-vector-based semantic similarity, but instead of one fixed cutoff, the merge threshold adjusts based on how important the skills are and how much their wording overlaps, wherein the system raises the threshold up to 0.97 for stricter matching when the skills involved are of high importance or share little lexical overlap, so that it does not carelessly merge skills that only sound similar but are not really the same.

Matching phrases are grouped under a single "master" skill, chosen as the most frequently used variant.

Example: *“AutoCAD” and “AutoCAD 2D and 3D” are merged into a single master skill, “AutoCAD”, since one already covers the other. “Electrical Troubleshooting” and “Maintenance” are kept separate because they appear in different contexts and do not clear the merge threshold.*

Step 2.7: Category Assignment

Lastly, each validated skill is mapped into a category using the system's learned skill dictionary, or is inferred from context if it is a new skill that the dictionary has not encountered before. Categorizing the skills makes it easier for the Recommendation Engine to match employees to project requirements later on.

Example: *AutoCAD → “Design Tools”; PLC Programming → “Programming”; Siemens S7 → “Industrial Automation”; Electrical Troubleshooting → “Maintenance”; Maintenance → “Maintenance”. The employee's final, structured skill profile is now ready to be used by the Recommendation Engine.*

Recommendation Engine

Once the employee skills are extracted and the Project Manager submits the project requirements, the Recommendation Engine computes and ranks the

candidates for the project, and the Resource Manager receives this ranked list before making the final assignment decision. The engine scores each employee using four criteria: Skill Matching, Workload, Availability, and Historical Performance. The first three criteria describe how well the employee currently fits the project, while Historical Performance brings in the employee's track record from past projects, so that consistently reliable employees are favored over those with only a good skill match on paper.

Step 1: Skill Matching

The system compares the employee's skill profile against the list of skills the Project Manager entered for the project, and counts how many of the required skills the employee actually has. Each required skill is also given a type by the Project Manager, either Primary or Secondary, so the system can weight skills that is core to the project higher than skills that is just nice-to-have. A Primary skill is given a weight of four (4), while a Secondary skill is given a weight of one (1), this is to make sure an employee who is missing a Primary skill is penalize more heavily than one who is only missing a Secondary skill, even if both employee matched the same number of skills overall. This produces a Matching Score between 0 and 1 that reflects what portion of the project's weighted required skills the employee covers.

$$\text{Eq. (1) Matching Score} = (\sum \text{weight of matched required skills}) / (\sum \text{weight of all required skills})$$

Example: A project requires PLC Programming (primary), Siemens S7 (primary), and Electrical Troubleshooting (secondary). Total weight = 4 + 4 + 1 = 9. An employee whose profile shows PLC Programming and Electrical Troubleshooting, but not Siemens S7, has a Matching Score of $(4 + 1) / 9 = 0.556$.

Step 1b: Prerequisite Skill Check

Because a high Matching Score could still be reached even if an employee is missing most of the project's Primary skills, as long as they cover enough Secondary skills to make up for it, the system applies a separate check before the final ranking is produced. The system computes what portion of the Primary skills specifically the employee has matched, this is called the Prerequisite Fulfillment. If an employee's Prerequisite Fulfillment is below fifty percent (50%), meaning the employee is matched less than half of the project's Primary skills, the employee is flagged as "Missing Core Skills," regardless of what their overall Recommendation Score is computed to be. This is to make sure an employee who is strong on optional skills but is weak on the project's actual core requirements is not recommended ahead of someone who is a better fit on the requirements that matter the most.

Eq. (1b) Prerequisite Fulfillment = (number of Primary required skills the employee has) / (total number of Primary required skills)

Example: A project requires four Primary skills and two Secondary skills. An employee who has only one of the four Primary skills, but both Secondary

skills, has a Prerequisite Fulfillment of $1 / 4 = 0.25$, which is below the 50% cutoff, so this employee is flag as "Missing Core Skills" and is not eligible for the normal "Strongly Recommended" / "Recommended" / "Consider" status tiers, even if their weighted Recommendation Score from Eq. (5) would otherwise place them highly.

Step 2: Workload Assessment

Instead of counting how many projects an employee is currently in, the system adds up the weight of every task that is currently active under that employee. Each task is given a weight based on the Priority the Project Manager sets when assigning it: Low priority is worth 1, Medium priority is worth 2, and High priority is worth 3. The Workload Score is simply the total of these weights, so an employee holding two heavy, high-priority tasks can register a higher Workload Score than an employee holding several low-priority ones.

Eq. (2) Workload Score $W = \Sigma$ (priority weight of each active task currently assigned to the employee)

Example: *An employee who is currently holding three active tasks rated High (3), Medium (2), and Low (1) priority has a Workload Score of $W = 3 + 2 + 1 = 6$.*

Step 3: Availability Calculation

The Workload Score from the previous step is converted into an Availability Factor between 0 and 1 using a sigmoid function, so that availability is

decrease smoothly as the workload increase, instead of dropping in a straight line. This approach of using a sigmoid function to avoid abrupt or radical changes in a score is also support by recent studies on optimization and resource allocation algorithms, wherein a sigmoid-based calculation is used specifically to prevent sudden jumps in a value as conditions change, and instead allow the value to transition smoothly (Du & Chen, 2025). This is design so a highly-loaded employee is not immediately reduced to zero, since this employee could still be the best or only available option for a certain project.

$$\text{Eq. (3) Availability Factor } A = 1 / (1 + e^{(0.4 \times (W - 7))})$$

***Example:** An employee with a Workload Score of $W = 6$ has an Availability Factor of $A = 1 / (1 + e^{(0.4 \times (6-7))}) = 1 / (1 + e^{(-0.4)}) = 1 / 1.670 = 0.599$, meaning the employee is already carrying a considerable load but is not treated as completely unavailable.*

Step 4: Historical Performance Evaluation

The Client provides two distinct outputs once a project is completed: a numerical Performance Rating and a written Project Deliverables Feedback. The Performance Rating, on a scale of 1 to 5 stars, is the value the Recommendation Engine converts into the Historical Performance Factor, while the Project Deliverables Feedback is stored alongside it in the employee's portfolio as supporting context for whoever reviews the recommendation later on. New employees who do not yet have a prior project history are given a neutral default value until real performance data becomes available.

Eq. (4) Historical Performance Factor HP = average_client_rating / 5

(default HP = 0.70 if no rating history exists yet)

Example: *An employee with an average client rating of 4.5 out of 5 stars across past projects has $HP = 4.5 / 5 = 0.90$, while an employee averaging 3.0 stars has $HP = 3.0 / 5 = 0.60$.*

Step 5: Combined Recommendation Score

The three factors above are combine using a weighted sum to produce the employee's final Recommendation Score, this is to make sure that skill match carries the most weight at 60%, availability is contribute 25%, and the historical performance is contribute the remaining 15%. Unlike a multiplicative approach, this is prevent the employee's score from being automatically push down to near zero just because one factor happens to be low.

Eq. (5) Recommendation Score Re = $(0.60 \times \text{Matching Score}) + (0.25 \times \text{Availability Factor}) + (0.15 \times \text{Historical Performance Factor})$

Worked Example: Matching Candidates to a Project

Project: *PLC System Redesign*

Required Skills: PLC Programming, Siemens S7, Electrical Troubleshooting

Table 1. Candidate Information

Candidate	Skills	Active Tasks	Average Client Rating
John Dela Cruz	AutoCAD, PLC Programming, Siemens S7	2 tasks (Medium, Medium)	4.5/5
Maria Santos	PLC Programming, Maintenance	3 tasks (High, Medium, Low)	3.0/5
Robert Tan	Siemens S7, Electrical Troubleshooting	1 task (High)	4.8/5

Table 2. Candidate Matching and Availability

Candidate	Skills Matched	Matching Score	Prerequisite Fulfillment (Eq. 1b)	Workload	Availability
John Dela Cruz	2/3	$8/12 = 0.667$	$2/3 = 0.667$	4	0.769
Maria Santos	1/3	$4/12 = 0.333$	$1/3 = 0.333$	6	0.599
Robert Tan	2/3	$8/12 = 0.667$	$2/3 = 0.667$	3	0.832

Table 3. Historical Performance and Recommendation Score

Candidate	Historical Performance	Recommendation Formula	Re
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John Dela Cruz	0.90	$(0.60 \times 0.667) + (0.25 \times 0.769) + (0.15 \times 0.90)$	0.727
Maria Santos	0.60	$(0.60 \times 0.333) + (0.25 \times 0.599) + (0.15 \times 0.60)$	0.440
Robert Tan	0.96	$(0.60 \times 0.667) + (0.25 \times 0.832) + (0.15 \times 0.96)$	0.752

Table 4. Final Ranking

Rank	Candidate	Re	Status
1	Robert Tan	0.752	Eligible
2	John Dela Cruz	0.727	Eligible
3	Maria Santos	0.440	Missing Core Skills

Robert Tan and John Dela Cruz are both recommend for the project, while Maria Santos is rank third with a Recommendation Score of 0.440. She is flag as "Missing Core Skills" because her Prerequisite Fulfillment is only $1 / 3 = 0.333$, meaning she is matched less than half of the project's Primary skills, which is below the fifty percent (50%) cutoff of Step 1b. Because of this, she is not eligible for the normal "Strongly Recommended", "Recommended", "Consider" status tiers, regardless of what her Recommendation Score is compute to be. This example also show why Availability and Historical Performance still matter even when two candidates tie on raw skill match. John Dela Cruz and Robert Tan both

cover 2 out of 3 required skills, but Robert Tan has a lighter workload and a slightly higher client rating, so his overall score is pushed above John's. This is to make sure the system does not just favor whoever looks equally qualified on paper, but the one who can deliver reliably given the current workload.

Continuous Learning and Feedback Loop

The loop does not end once an employee is deployed. Once an employee is assigned, the Project Manager receives the assigned employee and hands over the assigned task, and the employee then submits a weekly report while the project is ongoing, which keeps the workload data current. After the project wraps up, the Client sends back both a Performance Rating and a Project Deliverables Feedback, which are stored and used to update the employee's performance record and portfolio. This updated portfolio data is then fed back into the Recommendation Engine, so that future recommendations reflect real, up-to-date performance rather than a static profile.

Step 1: Performance Feedback Collection

After a project is completed, the Client rates the employee's performance on a scale of 1 to 5 stars, and separately submits written feedback on the project deliverables. Both are captured through the Client Feedback stage, and the manager may add additional notes on skill accuracy or workload suitability.

Example: A client gives John Dela Cruz a 5-star Performance Rating and adds Project Deliverables Feedback noting that the PLC panel redesign was delivered ahead of schedule.

Step 2: Portfolio Update

The employee's performance record and portfolio are updated with the new rating, and this updated Historical Performance value is what the Recommendation Engine uses the next time this employee is considered as a candidate, completing the feedback loop.

Software Engineering Paradigm

Rapid Application Development (RAD) Methodology

This study will use the Rapid Application Development (RAD) methodology to create OCR and NLP based resource recommendation system. RAD is iterative approach to focus on quick functional prototypes, regular user feedback, and flexibility in changing requirements. This methodology is good for this project because the OCR and NLP components need repeated testing and improvement using document and user input (Brill, 2025). This study follow phases:

Phase 1: Requirements Planning

In requirements planning phase, will identify the types of documents that system will be process using OCR and NLP. Also determining the information that will be extract from the uploaded documents. The development team should understand the system's expected functionality and objectives.

Phase 2: User Design

In user design phase, functional prototypes are develop for OCR text extraction, resource recommendation dashboards and document upload interface. The functional prototype will evaluate by respondents and review by the project requirements.

Phase 3: Rapid Construction

Iteration 1: OCR and Document Processing

The first iteration focus in developing the OCR component that extract texts from documents. The system will be testing using sample documents to ensure accurate and reliable text extraction. Every errors may found during the testing will be fixed before moving to the next iteration

Iteration 2: NLP and Information Extraction

In the second iteration, Natural Language Processing (NLP) will be applied to the extracted text to identify and extract relevant information for resource recommendation. The extracted text will undergo preprocessing, including text cleaning, tokenization, normalization, and synonym mapping. Candidate skill phrases will then be validated using a Machine Learning

classification model based on Term Frequency–Inverse Document Frequency (TF-IDF) and Logistic Regression to determine whether they represent valid employee skills before being stored in the structured employee profile. When the trained classification model is unavailable or there is insufficient human-verified training data, the system will temporarily apply a rule-based fallback mechanism to continue validating candidate skills. The classification model will be continuously improved through user-verified feedback, allowing it to be retrained and enhance its accuracy over time. Through repeated testing using sample documents, this iterative process will ensure that the system correctly extracts, validates, and organizes employee skills for the recommendation process.

Iteration 3: Resource Matching and Recommendation Algorithms

In the last iteration, algorithms will be developed to match extracted resources with project needs and generate recommendations. The current baseline recommendation process will use set-intersection skill matching to compare employee skills with project requirements and generate initial recommendations. The algorithms will be tested using sample data to verify their accuracy, relevance, and efficiency to determine whether they provide meaningful suggestions.

Phase 4: Cutover

After all the iterations, the system parts are put together and tested to see if they work well. The system will be checked using test cases and sample data to make sure it works correctly and performs okay before it is used.

Benefits of RAD

Using RAD helps the system faster because working functional prototypes can be developed quickly. Feedback from test cases and internal evaluations helps improve the system. This allows the NLP and recommendation algorithms to be changed based on the results from early tests with this the system more accurate and reliable.

Population and Sampling

The target population of this study have key personnel that are in workforce allocation and technical operation within WEA and other companies that have the same work position. It includes the Resource Managers that are responsible for assigning employees to projects, Project Managers who decide the specific manpower and safety requirements for each hazardous industrial sites and the Skilled Employees whose qualifications serves as the primary data inputs of the system, including also the Project Managers, Resource Managers, Employees and HR personnel from other companies that have the same work position. Focusing in this roles, the study make sure that the Skill-Extraction and

Workload-Based Resource Recommendation System is reflects the actual operation processes and challenges that encountered in managing construction personnel. The study use purposive sampling, a technique that used to select participants based on their specific relevance and expertise. This is appropriate because the evaluation of the system is requires respondents with specialized knowledge in workforce allocation, construction project management and software development. According to Andrade (2021), purposive sampling is effective when researchers intentionally choose participants who possess the qualifications and characteristics that are necessary to address the study's objectives.

Total of seventeen (17) are purposively selected as the respondents who will evaluate the system based on the ISO/IEC 25010 software quality standard, coming from both WEA and other companies with the same work positions. This group is consist one (1) domain expert with experience in workforce documentation and validation to verify the business logic, three (3) technical experts that will assess the system's architecture and the integration of the OCR and NLP components and lastly the thirteen (13) end users who will evaluate the system usability and functionality within the WEA context and other companies. The Functional Suitability, Reliability and Usability is evaluate by the end-users, domain and IT/technical since this is the one experienced by them directly, whether the features is working, consistent, and easy to use. While the Performance Efficiency, Maintainability, Portability and Security is evaluate by the IT/technical experts since this are technical/code-level concerns that is not

actually seen or experienced by the end-user. This sample size is justified because it provides balanced perspective from the both technical developer and actual business people who will utilize the system.

Data Collection Methods

The primary instrument used in this study is a structured Likert-scale questionnaire that is based on the ISO/IEC 25010 international software quality standard. The questionnaire was used to gather the evaluations and assessments of the seventeen (17) purposively selected respondents, which consists of thirteen (13) end users, three (3) technical experts, and one (1) domain expert. All seventeen (17) respondents, which is the end users, domain expert, and IT/technical experts, answer the Functional Suitability, Reliability, and Usability sections of the questionnaire, since this are the quality characteristic that they experienced directly while using the system, such as whether the features is working, consistent, and easy to use. Only the three (3) technical experts, meanwhile, answer the remaining sections, which are Performance Efficiency, Maintainability, Portability, and Security, because this are technical and code-level concerns that is not actually visible or experienced by the end-user or domain expert, such as how maintainable the system is, how it perform under load, or how secure the implementation is, which only the IT/technical experts are qualified to judge (Lagman et al., 2025). It is design to assess the quality of the Skill-Extraction and Workload-Based Resource Recommendation System in

terms of the seven main quality characteristics of the system as set by ISO/IEC 25010. Each item or question in the questionnaire corresponds to a specific quality characteristic and is rated using a Likert scale so that the respondents are able to express the degree of their agreement on how the system performs.

The scale ranges from 1 to 5, where 1 means Strongly Disagree and 5 means Strongly Agree, this is the way of allowing the evaluators to clearly rate the system quality characteristic based on their actual experience in using the system. The questionnaire is divided into sections that cover the seven main quality areas of the study, which are Functional Suitability, Reliability, Usability, Performance Efficiency, Maintainability, Portability, and Security, this aligns with the ISO/IEC 25010 model that is used as the basis of the evaluation. The functional suitability section checks whether the system is able to correctly and function properly to extract skills from the employee internal resume, analyze the workload distribution, and produce accurate and fair recommendations of employee to their specific project. It also includes indicators that are directly aligned with the research questions of the study. The first research question focuses on whether the overall system can be developed for WEA to automate skill extraction and support the intelligent resource allocation, so items were included to evaluate the system's core functionalities as a whole. The second research question addresses how the system extracts and classifies the employee skills from the internal resume documents through OCR and NLP, so additional items were included to evaluate if the system can accurately convert and categorize the skill information that is contained in the internal resume. The third research question

addresses how the system analyze the workload distribution across employees to identify who is underutilize or overload, so an indicator was included to measure whether the system correctly show the current workload status of each employee. The fourth research question addresses the effectiveness of the recommendation engine in generating accurate and fair recommendations based on the skill matching, availability, and workload conditions, so an indicator measuring the system's ability to show the most suitable and available employees who are not currently assigned to any project was also added. Additional indicators were also included to evaluate the Human Resources module, particularly on how the system present the applicant's uploaded resume alongside their manually input details, so the Human Resources personnel are able to check and manually filter and recommend the applicant based on this comparison.

The usability section evaluates if the interface is easy to navigate and understand for each type of user, which are the Resource Manager, Project Manager, Human Resources and Employee. And the reliability section assesses if the system performs consistently and without errors during the normal use, and is able to recover properly when an error or interruption occurs. This aligned with the seventh research question that focuses on the level of acceptability of the system in terms of functional suitability, reliability and usability. The remaining sections, particularly Performance Efficiency, Maintainability, Portability and Security, is focus more on the technical and code-level aspect of the system, such as how fast the system process and return the OCR/NLP skill extraction

results and the ranked recommendation list, how the individual modules like the skill extraction and recommendation engine can be identify, diagnose and update without affecting the whole system, how the system function properly across different browser and device, and how the system restrict access base on user role and protect the account through session timeout and lockout after failed login attempts. By using this instrument, the researchers can gather a structured and measurable data that will be used to determine how well the Skill-Extraction and Workload-Based Resource Recommendation System meets the quality standards that is set by ISO/IEC 25010.

Pamantasan ng Lungsod ng Pasig – College of Computer Studies
Resource Recommendation System for WEA
 ISO/IEC 25010 Software Quality Evaluation Questionnaire

Please rate each item based on your experience using the system. Use the scale below:

1 — Strongly Disagree 2 — Disagree 3 — Neutral 4 — Agree 5 — Strongly Agree

A. Functional Suitability					
Indicator	1	2	3	4	5
1. The system accurately extracts and categorizes skills from the uploaded internal resume (CV).	☐	☐	☐	☐	☐
2. The system correctly matches employee skills and historical performance to the specific requirements of a project.	☐	☐	☐	☐	☐
3. The system correctly shows the availability of employees who are not currently assigned to any project.	☐	☐	☐	☐	☐
4. The system correctly displays the applicant's uploaded resume alongside their manually input details for comparison.	☐	☐	☐	☐	☐
5. The system's generated recommendations are relevant and useful for making assignment decisions.	☐	☐	☐	☐	☐
6. The system's recommendations improve or adjust appropriately based on an employee's past project performance ratings.	☐	☐	☐	☐	☐
7. The system's dashboards and reports provide useful insights for identifying underutilized or overloaded employees, workforce capacity, and skill gaps.	☐	☐	☐	☐	☐
B. Reliability					
8. The system performs consistently without errors during normal use.	☐	☐	☐	☐	☐
9. The system is able to recover properly (e.g., no data loss) if an error or interruption occurs.	☐	☐	☐	☐	☐
10. The skill extraction and recommendation results remain consistent across repeated use with the same data.	☐	☐	☐	☐	☐
C. Usability					
11. The interface is easy to navigate and understand for my specific role (Resource Manager / Project Manager / HR / Employee).	☐	☐	☐	☐	☐
12. I was able to complete my intended task (e.g., uploading a resume, reviewing a recommendation) without needing outside help.	☐	☐	☐	☐	☐
13. The system provides clear feedback or explanation on how a recommendation was generated.	☐	☐	☐	☐	☐
D. Performance Efficiency					
14. The system processes and returns the OCR/NLP skill extraction results within an acceptable amount of time.	☐	☐	☐	☐	☐
15. The system generates the ranked recommendation list without noticeable delay.	☐	☐	☐	☐	☐
16. The system's dashboards and reports load and update in a reasonable amount of time.	☐	☐	☐	☐	☐
E. Maintainability					
17. Errors or issues in the system's modules (e.g., OCR, NLP, recommendation engine) can be easily identified and diagnosed.	☐	☐	☐	☐	☐
18. Individual components of the system (e.g., skill extraction, recommendation engine) can be tested or updated without affecting the entire system.	☐	☐	☐	☐	☐
F. Portability					
19. The system is accessible and functions properly through a standard web browser without needing special installation.	☐	☐	☐	☐	☐
20. The system's core functions still work as expected across different devices or browsers used for access.	☐	☐	☐	☐	☐
G. Security					
21. The system properly restricts access based on user role (Admin, Resource Manager, Project Manager, HR, Employee).	☐	☐	☐	☐	☐
22. The system's login and account protection features (e.g., lockout after failed attempts, session timeout) work as intended.	☐	☐	☐	☐	☐

Comments / Suggestions (optional):

Figure 3. ISO/IEC 25010 Evaluation Questionnaire

Figure 3 provides the design of the questionnaire used in this study. It is designed specifically to gather the structured assessment of the evaluators on

the different quality characteristics of the system based on the ISO/IEC 25010 standard. This same questionnaire is used as the instrument to collect all the data needed for evaluating the overall quality of the Skill-Extraction and Workload-Based Resource Recommendation System.

The data collection started with preparing the ISO/IEC 25010-based Likert-scale questionnaire that will be used by the evaluators in assessing the system. After completing the system functional prototype, the researchers let the purposively selected respondents use and interact with the features of the Skill-Extraction and Workload-Based Resource Recommendation System. The evaluators were given time to explore the system by performing tasks that are relevant to their role, such as uploading employee internal resume through the Employee Portal, viewing the ranked employee recommendations on the Edit Project and Assign Team screen, checking how the system match the internal Resume/CV to the project requirements, and for the Human Resources personnel, checking the applicant's uploaded resume against their manually input details on the Applicant Filtering screen before deciding who to filter and recommend. After using the system, the evaluators answered the ISO/IEC 25010-based Likert-scale questionnaire to rate the system based on their experience and assessment. The responses were then collected and compiled by the researchers for data analysis.

Data Analysis Procedures

This section will describe how the collected data is going to be processed, analyzed and interpreted to support both system's automated recommendations and evaluation of effectiveness. The process combines of the structure data transformation, the system performance, analytics modeling and the respondent based evaluation that will ensure the assessment of the Skill-Extraction and Workload-Based Resource Recommendation System for WEA.

Data Processing and Preparation

The system will start by collecting the documents of the employee internal CV, and using Optical Character Recognition or OCR to convert them into a system readable text and the data that is extracted in employee credential will undergoes preprocessing that includes the cleaning of unnecessary characters, normalizing of terms and removing the duplicates or informations that are irrelevant to the system. After that, the Natural Language Processing or NLP techniques are applied to the extracted data such as skills in internal CV, which are then stored in a database for an efficient analysis. For the applicant side, the resume that is upload together with the manually input application details are also stored in the database, so that the Human Resources personnel can later compare both when reviewing an applicant.

Skill Analysis and Structuring

The extracted data will be analyzed using Natural Language Processing or NLP techniques such as keyword matching, tokenization and synonym

mapping to convert the different variations of skills into a consistent terms. Candidate skill phrases will then be scored by a trained TF-IDF and Logistic Regression classifier to determine whether they represent valid skills before being confirmed into the structured skill profile. The system will then organize the informations that are already processed into structured datasets, that includes the skill profiles, project requirement and historical performance classifications of the employees. This structured representation are enables accurate and efficient comparison between the employee qualifications and the specific needs of the project.

Matching and Recommendation Analysis

The resource manager will accept the project and the recommendation will start the system to perform matching analysis by comparing the employee profiles with the project requirements using mathematical models. It will determine the skill overlap, computes matching scores and ranks the employees based on their suitability to the project. Filtering mechanism will remove the candidates with no relevant skills to the project, while the allocation analysis calculates the workload score of each employee based on the priority weight of their active tasks and converts this into an availability factor. The client's historical performance rating is also factored in as part of the final computation, so that the recommendation score reflects skill matching, availability and historical performance together rather than skill match alone. These processes will ensure that only the employees that are qualified will be recommended and that the resources are distributed efficiently to the project.

Performance and Accuracy Analysis

To evaluate the effectiveness of the system, a quantitative analysis will be conducted using accuracy measure for both Optical Character Recognition or OCR and Natural Language Processing or NLP components. The OCR accuracy is determined by comparing the extracted text with the original document content of the employee, while the NLP accuracy will be evaluated by comparing the extracted skills with the manually identified data. The recommendation accuracy will be also assessed by verifying whether the suggested employees are meets the project requirements of the project.

The formula used to compute OCR performance is based on Word Error Rate (WER) and word-level accuracy. WER is computed as $WER = (S + D + I) / N$, where S is the number of substituted words, D is the number of deleted (missing) words, I is the number of inserted words that the OCR output produced but that do not appear in the ground-truth text, and N is the total number of words in the ground-truth text. Word-level accuracy is computed separately as $Accuracy = ((N - S - D) / N) \times 100$. Because insertion errors are counted in the WER calculation but are not subtracted in the accuracy calculation, these two metrics are not strict complements of one another, a document with a higher number of OCR-inserted extraneous words can therefore show a WER and an accuracy percentage that do not sum exactly to 100%. For the NLP skill extraction and ML skill classification components, performance is evaluated using a confusion matrix, from which Precision, Recall, and F1-score are derived to measure how accurately the system identifies and classifies valid employee skills.

The OCR component was evaluated on eleven (11) synthetic internal CV documents. Since the files were digitally generated PDFs, each was converted into an image-only file so that the system would use OCR. Each document was tested under four (4) conditions: a baseline (converted to an image with no degradation) and three simulated scan-quality levels (easy, medium and hard), which apply random skew, blur, noise, brightness and contrast changes, and JPEG compression in increasing strength. This produced forty-four (44) scans, and the degradation was seeded so that the test can be repeated. The OCR output was compared with a manually transcribed ground truth of 4,849 words per condition, and the S, D and I counts were combined across the documents within each condition.

The NLP skill extraction was also measured on the same scans. A human reader listed the 191 skills found in the CVs, and a skill was counted as found if the system surfaced it as a candidate skill, whether auto-approved or set aside for the Employee's review. The recall is computed as $(\text{skills found} / \text{skills expected}) \times 100$. A skill was matched if the phrases were identical, or if one phrase contained the other and the shorter one covered at least half of the words of the longer one. This measures only whether the skills were extracted, not whether the other candidates were correct.

System-Based Data Analysis

The system based analysis will be focus on the overall performance of the automated components including the OCR, NLP and the recommendation

engine. The accuracy of the system processes the data, the generates matching the score and the produces ranked recommendation will be analyze by the researchers. In evaluating the consistency and reliability, the quantitative metrics such as accuracy percentage, matching scores and allocation computation will be used. The result will be organize and will be process using the tools such as Python and Supabase to ensure the systematic tracking and validation of the output.

Respondent-Based Evaluation Analysis

The respondent based evaluation will be focusing on the assessing the system from the perspective of the selected evaluations using the ISO/IES 25010 based likert-scale questionnaire. By computing the mean scores for functional suitability, reliability and usability, Performance Efficiency, Maintainability, Portability and Security the responses will be analyze quantitatively. With these scores will provide the clear measurement of the system quality based on the user experience.

Considering a feedback and observations from the respondents during system interaction, the qualitative interpretation is being incorporated. This will help to identify the usability concern and what areas needed to improve. The researchers will obtain balance evaluation both technical performance and user satisfaction by combining the quantitative ratings and the qualitative insights.

Ethical Considerations

Data Privacy and Confidentiality

The researchers ensures that all employee data that are extracted through OCR and processed through the NLP component is stored securely at all times. This also cover the applicant's uploaded resume and their manually input application details, as well as the client's performance rating and feedback, since these are also considered sensitive information that the system collects and store. Personal information are encrypted both in the database and during transmission to prevent unauthorized access or data breaches. Access controls are also implemented so that only the authorized users can view or manage the sensitive information, that ensure the privacy of the employees and applicants whose documents are being process by the system.

Fairness and Bias Reduction

The recommendation algorithm of the system is designed to give fair and unbiased suggestion to the Resource Manager. Skill matching, workload, availability and historical performance are based on the actual qualifications, current tasks and verified client feedback of the employees, not on personal characteristics or other non-relevant factors. By doing this approach, equal opportunities are given to all the employees so that the system will not favor or overlook anyone, which addresses the issue of underutilization that were identify in this study. Since the classifier is learn from the human-provided labels rather than fixed rules, the employees can review and correct their own extracted skills

through the system's feedback feature, this is to help make sure that labeling errors will not unintentionally cause the system to favor or disadvantage certain employees. On the Resource Manager's side, fairness is maintained by keeping their role limited to matching the verified skills of employees to project requirements, so their judgment is not the source of bias in the recommendation. On the Human Resources side, fairness is also maintain by having the applicant's resume and manually input details shown together, so the filtering and recommendation decision is based on comparing actual submitted information rather than personal preference.

Transparency

The system is designed to provide clear explanations for every recommendations that it generates so the users can understand how the decisions are being made. The Resource Manager can see the basis of each recommendation including the matched skills, the availability, and the historical performance of the employees. A documentation is also provided for all system functions to make sure everything is clear and understandable to the users.

Security

To protect the system from any unauthorized access, user authentication is implemented using JSON Web Tokens (JWT) so that only the authorized

persons can use and access the system. Session management is also applied, wherein the system automatically log out the user after a set period of inactivity to prevent unauthorized use especially when the user forgot to log out. To further prevent unauthorized access, the system locks the user account after several failed login attempts, this can be unlock manually by the Admin. The system also implements role-based access control, this is to ensure that each user, whether Admin, Resource Manager, Project Manager, Human Resources or Employee, can only access the features and data that are relevant to their role. All the important actions in the system, such as login attempts and account locking, are recorded through audit logging for accountability and monitoring purposes. Lastly, the system also provides a secure password reset process through a time-limited link that are sent to the user's email, this is to prevent unauthorized password changes.

Academic Integrity

The researchers made sure that all the sources and references that are used in this study is properly cited and acknowledged following the APA citation format. The research are also follows the established development and documentation standards to maintains the quality and standard of the study. The testing results are reported accurately and without any manipulation to ensure the honesty and integrity of the findings of the study.

This chapter described the research design, the architectural framework, the algorithm and the software engineering paradigm that guide the development of the Automated Skill Extraction and Workload-Based Resource Recommendation

System for WEA, together with the population and sampling, the data collection methods, the data analysis procedures and the ethical considerations that are observe throughout the conduct of this study. The following chapter presents the results that are gather from the testing and evaluation of the system, together with the discussion of these results in relation to the objectives and the research questions that are raised in Chapter 1.

CHAPTER 3

RESULTS AND DISCUSSIONS

Introduction

This chapter presents the results that are gathered from the development, implementation and evaluation of the Automated Skill Extraction and Workload-Based Resource Recommendation System for WEA. This study aims to develop an automated skill extraction and workload-based resource recommendation system for WEA to improve workforce allocation, task matching and operational efficiency. The results in this chapter are organized according to the Statement of the Problem that is present in Chapter 1, and it begins with the discussion of the project capabilities and the testing process that were conducted to determine whether the system is functioning as intended, followed by the presentation of the testing results that were gathered from the functional, technical and acceptability evaluation of the system.

Project Capabilities and Testing Process Details

The Automated Skill Extraction and Workload-Based Resource Recommendation System is designed as a web-based platform that is built using React and Vite for the frontend, Node.js and Express for the backend, Python for the Optical Character Recognition (OCR) and Natural Language Processing

(NLP) components, and Supabase as the database and authentication service. The system is organized into different modules that correspond to the specific roles that are involved in the workforce allocation and hiring process of WEA, namely the Applicant, Employee, Human Resources, Project Manager, Resource Manager, Admin and Super Admin modules.

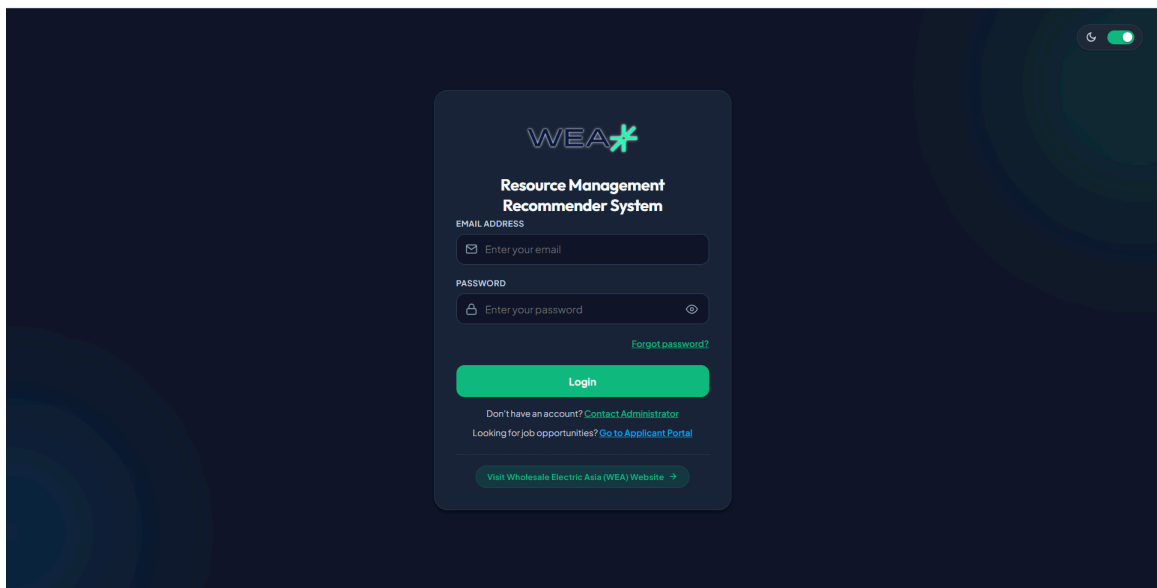


Figure 4. System Login Page

The system's core capability is address the first research question by providing an automated pipeline that extract, classify and validate employee skills from the internal resume of WEA, which is uses a standardized format for its employee documentation. The Skill Extraction module is compose of an OCR component that uses Tesseract OCR through Pytesseract to convert the uploaded internal CV, which is only accept the PDF, PNG and JPEG file formats, into a machine-readable text. The image is undergo grayscale conversion and adaptive thresholding using Otsu's method before it is process by the OCR

engine, in order to improve the readability of the document prior to text recognition.

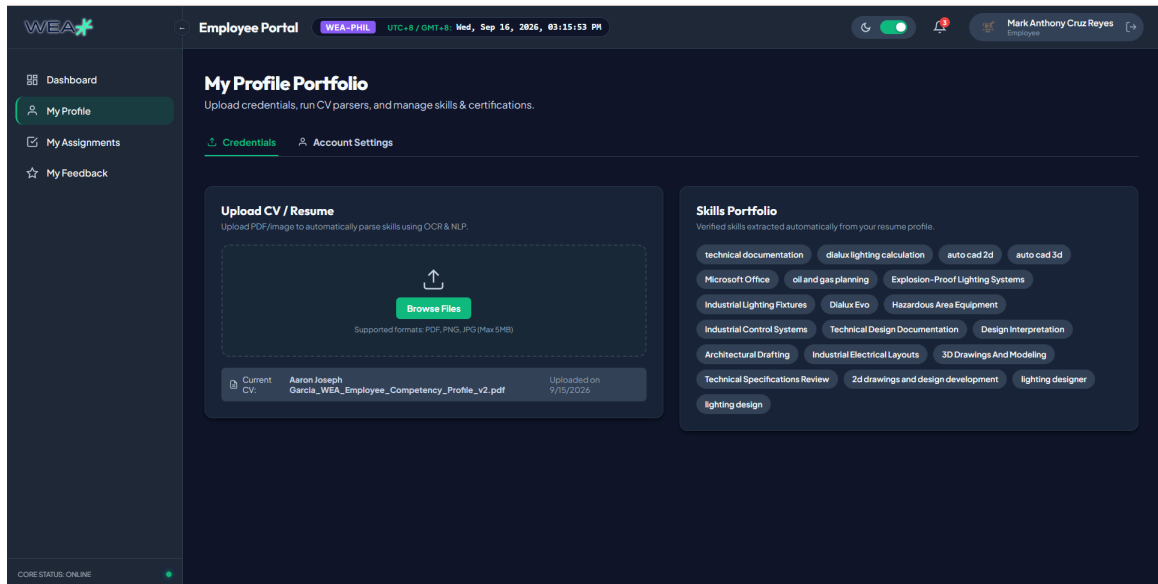


Figure 5. Employee Portal

Once the raw text is obtain, the NLP module is process the text through a pipeline that is compose of tokenization using the spaCy library, noun phrase extraction, and a Machine Learning based skill classification. The classification is perform using a TF-IDF feature representation that is combine with a Logistic Regression model that assigns a confidence score to every candidate phrase. The system is automatically approves a phrase as a skill when the confidence score is reach 0.75 or higher, while a phrase that scores between 0.50 and 0.75 is treated as a likely skill and is set aside for the Employee to review, and a phrase that scores below 0.50 is reject as not a skill. Any score that is fall between the reject and likely-skill range is mark as uncertain and is also surface for the Employee's manual confirmation. The phrase that the Employee is

approves or rejects through this review screen is stored as a labeled example, and once at least twenty (20) verified examples have accumulate, the classifier is retrain using this human-verified data, this is to allow the model's accuracy to improve as more employees interact with the system.

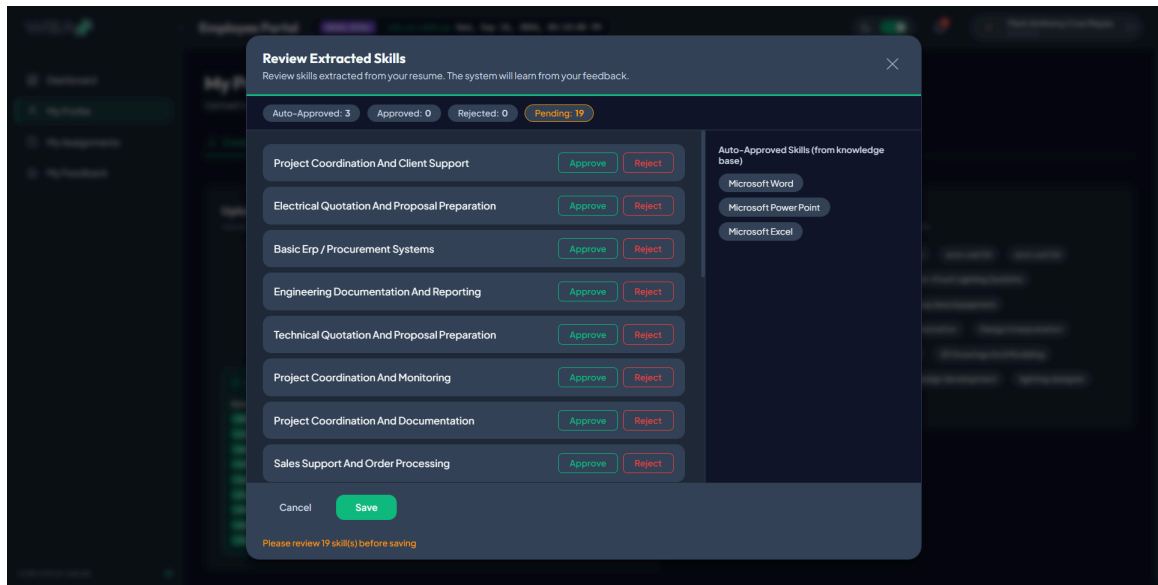


Figure 6. Skill Feedback / Skill Review screen

For the workload and resource recommendation capability that is address the third and fourth research questions, the system tracks each employee's active tasks and computes a Workload Score that is base on the priority weight of every task, wherein a Low priority task is weighted as one (1), a Medium priority task as two (2), and a High priority task as three (3). This Workload Score is then converted into an Availability Factor that is ranges from 0 to 1 using a sigmoid function, so that the availability is decrease smoothly as the workload increase instead of dropping in a straight line, and even an employee with a very high workload is never reduced all the way to zero, this is to make sure this employee

could still be consider if they are the best or only available option for a certain project. The Recommendation Engine of the system is combines this Availability Factor with the employee's Matching Score, which is reflects how many of the project's required skills the employee possesses, and the employee's Historical Performance Factor, which is derive from the average client rating that the employee received on completed projects and is default to a neutral value of 0.70 for an employee without any prior rating history. These three factors are combine using a weighted sum, wherein the Matching Score is contribute 60%, the Availability Factor is contribute 25%, and the Historical Performance Factor is contribute the remaining 15%, to produce the final Recommendation Score, which is use by the system to rank the candidates that are then present to the Resource Manager for the final assignment decision.

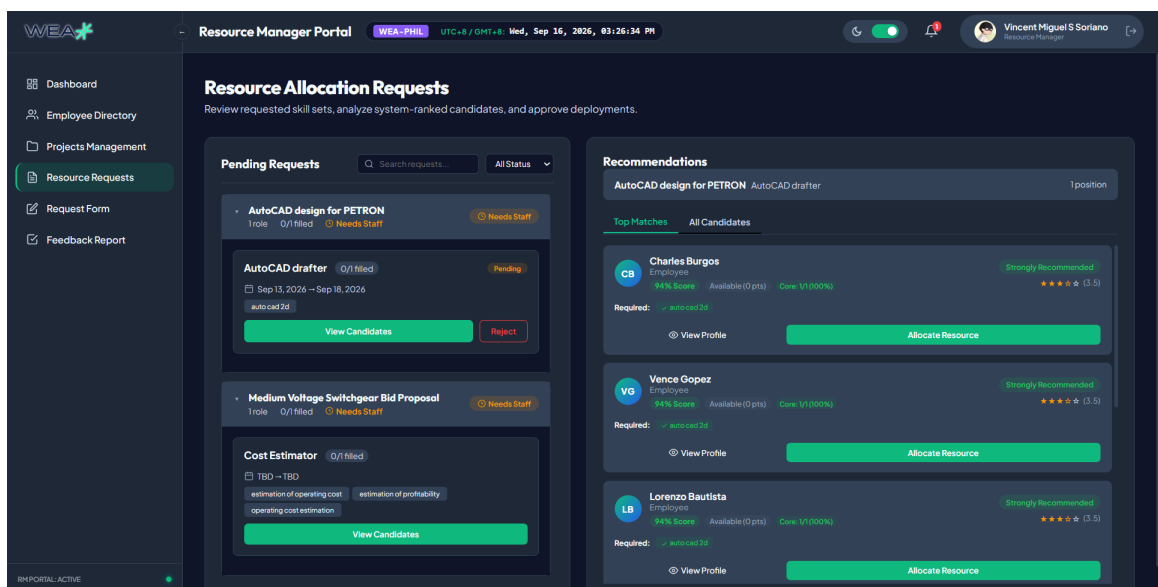


Figure 7. Resource Manager's Ranked Recommendation List

To address the fifth research question, the system is design to continuously learn and adapt from the outcomes of past assignments. After a project is complete, the Client is provides a Performance Rating and a Project Deliverables Feedback through the system, and this rating is updates the employee's Historical Performance Factor, which is then use by the Recommendation Engine the next time the same employee is consider for a new project assignment. This is creates a feedback loop wherein the recommendations of the system are inform by the employee's actual track record instead of relying only on a static skill profile.

WEA

Project Feedback

Hi Lynn Czyla Alpuerto, please rate each team member who worked on **Industrial Warehouse Lighting Design**. Your feedback helps WEA improve and helps the team grow.

Romell Ebuon
Project Manager

Overall Rating *

Technical Skills *

Communication *

Timeliness *

Quality of Work *

Teamwork *

Problem Solving *

Strengths

What did this person do particularly well?

Areas for Improvement

Any areas where this person could improve?

Would you recommend this person for future projects? *

Yes No

Figure 8. Client Feedback Form

For the sixth research question, the system is also integrate data analytics that provide insights into the workforce capacity of WEA. The Resource Manager modules is include dashboards that display the department-level workload and utilization, this is allowing the identification of employees or departments that are

underutilize or overload, together with the reports that support the strategic planning of the workforce.

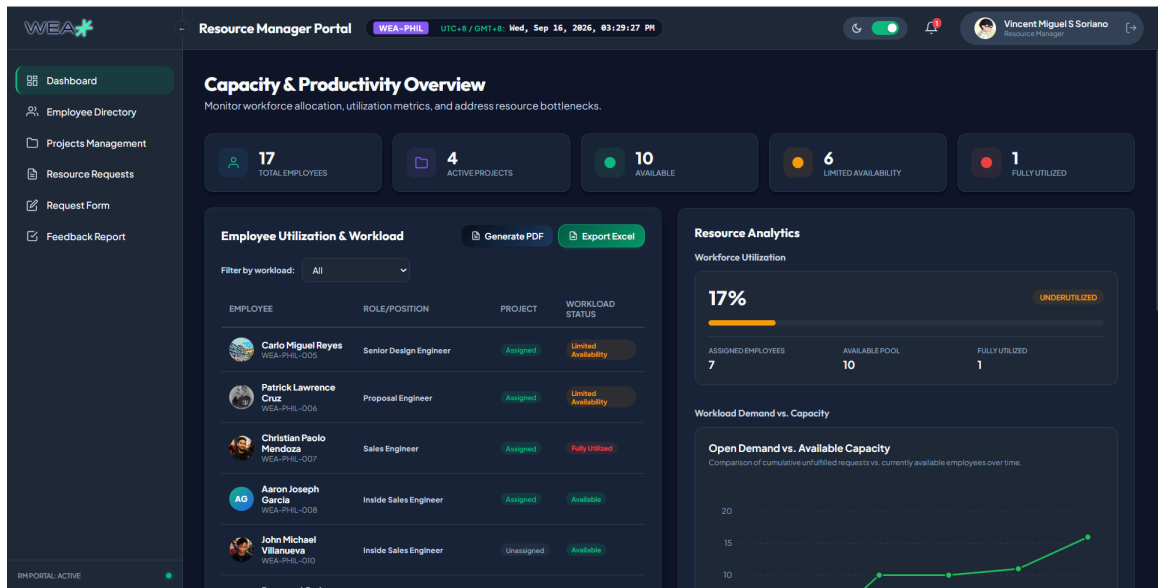


Figure 9. Resource Manager Analytics Dashboard

Aside from the internal workforce allocation, the system also cover the external hiring process of WEA through the Human Resources module, wherein an Applicant may submit an application form together with a resume through the Applicant Portal. Unlike the internal employee's internal CV, the applicant's resume is only serve as a supporting document and is not automatically process through OCR and NLP, since the Applicant is require to manually input their skills and other relevant details through the application form. This is allowing the Human Resources personnel to review and compare the applicant's uploaded resume against the manually input information before making the final hiring decision, and once an applicant is hire, they become part of the internal talent pool that the Resource Manager can consider for future project assignments.

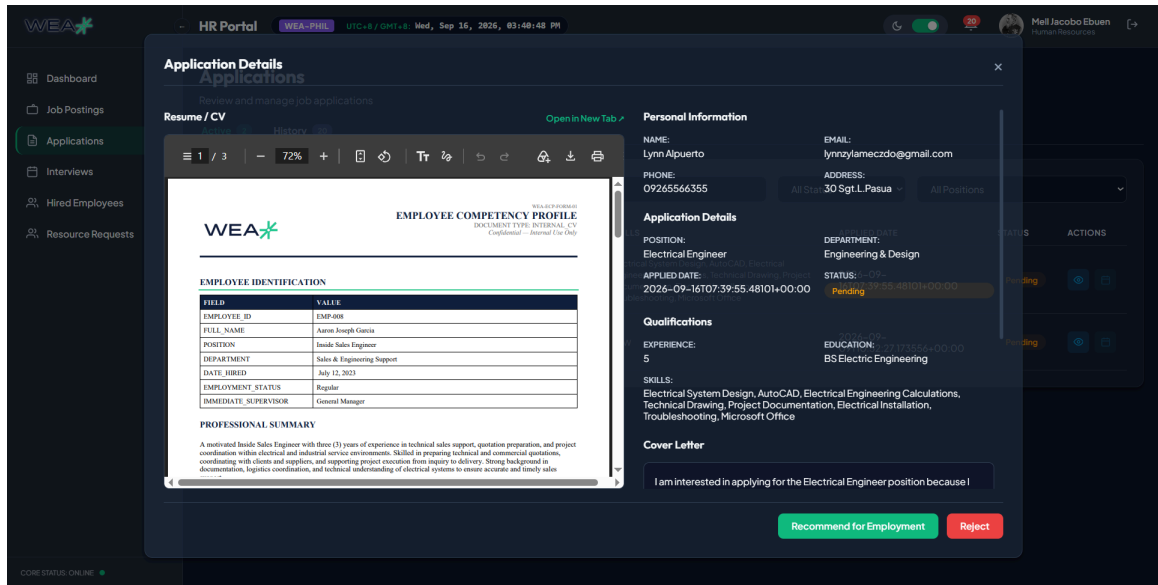


Figure 10. HR Applicant Filtering

To determine whether the system is correctly implements these capabilities, the researchers conduct a testing process that is cover the functional components of the system across all of the modules that are describe above. The strategy that is use for this testing process is the black-box approach, wherein each module is evaluate base on its expected input and output alone, without the tester need to examine the underlying source code. The objective of the testing is to verify whether each feature of the system, such as the document upload and processing, the skill review and confirmation, the resource recommendation, the workload analytics and the applicant filtering, is perform their intended function correctly. The resources that is use for this testing process is the researchers themselves as the tester, together with sample employee record, sample resume and sample project data that is prepare specifically

for the testing, this is to make sure that the actual data of WEA is not use or affect during the testing. The testing is follow a structured test plan, prepare and record in a separate Excel file, that lists the test scenario, the test steps, and the expected result for every test case per module, and this test plan is carry out across the span of the system's development and prior to its deployment for the evaluation. A test case is consider to pass if the actual result of the system is match the expected result that is state in the test plan, and is consider to fail if otherwise, this success criteria is apply consistently across all ten (10) modules of the system.

Functional testing of the system followed the black-box testing approach, in which each module is evaluated based on its expected inputs and outputs without examining the underlying source code (Nidhra & Dondeti, 2012).

Table 5. Summary of Functional Testing Results per Module

Module	Total Test Cases	Passed	Failed	Not Executed	Pass Rate (%)
Applicant Module	28	27	1	0	96.43
Authentication & Account Module	48	47	0	1	97.92
Admin Module	70	69	1	0	98.57
Human Resource Module	72	71	0	1	98.61
Employee Module	52	51	1	0	98.08

Project Manager Module	94	94	0	0	100.00
Resource Manager Module	50	50	0	0	100.00
Super Admin Module	68	65	0	3	95.59
Client Feedback Module	8	8	0	0	100.00
Shared Common Module	17	13	0	4	76.47
Overall Total	507	495	3	9	97.63

Base on the result that is gather from the Excel test plan, the functional testing is cover a total of five hundred seven (507) test cases across the ten (10) modules of the system, with four hundred ninety-five (495) test cases that is pass, three (3) that is fail, and nine (9) that is not yet execute, resulting to an overall pass rate of 97.63%. The Project Manager Module, Resource Manager Module, and Client Feedback Module obtain a perfect pass rate of 100.00%, which show that these modules' functions are working as intended base on the expected results.

On the other hand, the Applicant Module, Admin Module, and Employee Module each record one (1) failed test case. In the Applicant Module, the system is not display the "You have already applied for this position" message and

instead allow a duplicate application to be submitted, which mean the duplicate-prevention logic for the same job posting is not working as expected. In the Admin Module, the Role field is default to "Employee" instead of requiring the Admin to explicitly select a role, so the "This field must not be left blank" validation message is not trigger when the Role is left unselect. In the Employee Module, the system is not display the "Skill feedback saved successfully" message after the Employee accept or reject the extracted skills and click Save, which mean the save action either fail to complete or the confirmation message is not trigger as expected.

The Shared Common Module obtain the lowest pass rate among the ten (10) modules, with 76.47%, since four (4) of its seventeen (17) test cases, particularly those on viewing the status of a previously sent message and on the "Failed to update settings" error handling, is not yet execute. The Super Admin Module obtain the second lowest pass rate, with 95.59%, since three (3) of its sixty-eight (68) test cases, particularly those on assigning and removing a branch manager and on the dashboard's error handling, is not yet execute. The Authentication & Account Module and the Human Resource Module, meanwhile, each have one (1) test case that is not yet execute, which is the password reset error handling and the Admin notification for a new employee account, respectively.

Overall, the functional testing cover a total of five hundred seven (507) test cases, with four hundred ninety-five (495) that is pass, three (3) that is fail, and nine (9) that is not yet execute, resulting to an overall pass rate of 97.63%.

Aside from the functional testing, the researchers also conduct an accuracy evaluation for the automated components of the system, particularly the OCR and NLP-based skill extraction and the recommendation engine, this is to determine the reliability of the system's core algorithm before it is present to the respondents for the ISO/IEC 25010 based evaluation.

OCR extraction performance is measure using Word Error Rate (WER), its a standard metric that quantifies the word level substitutions, deletions, and insertions needed to align the extracted text with the ground-truth text (Sohail et al., 2024). Skill classification performance is measure using confusion matrix and the derived Precision, Recall and F1-score metrics, which are commonly used to evaluate classification models (Sujon et al., 2025).

Table 6. *OCR Text Extraction Results by Scan-Quality Condition*

Condition	Sample Size	Measure	Result
Baseline (no degradation)	11 internal CV documents (4,849 reference words)	Word-level accuracy (S = 676, D = 426, I = 369)	77.3% (WER = 30.3%)
Easy scans	11 internal CV documents	Word-level accuracy (S = 606, D = 468, I = 399)	77.9% (WER = 30.4%)

	(4,849 reference words)		
Medium scans	11 internal CV documents (4,849 reference words)	Word-level accuracy (S = 837, D = 531, I = 369)	71.8% (WER = 35.8%)
Hard scans	11 internal CV documents (4,849 reference words)	Word-level accuracy (S = 1,650, D = 446, I = 857)	56.8% (WER = 60.9%)
Total (all conditions)	44 scans (19,396 reference words)	Word-level accuracy (S = 3,769, D = 1,871, I = 1,994)	70.9% (WER = 39.4%)

As presented in Table 6, the OCR component was evaluated on eleven (11) internal CV documents (4,849 reference words per condition) under four (4) conditions. The word-level accuracy was 77.3% (WER = 30.3%) on the baseline, 77.9% (WER = 30.4%) on the easy scans, 71.8% (WER = 35.8%) on the medium scans and 56.8% (WER = 60.9%) on the hard scans, or 70.9% (WER = 39.4%) across all forty-four (44) scans. The difference between the baseline and the easy scans was 0.6 percentage points. From the baseline to the hard scans, the substitutions increased from 676 to 1,650 and the insertions from 369 to 857, and each of the eleven (11) documents had a lower accuracy at the hard level than at

its own baseline. The sample error cases of the baseline included words that were read in a different order from the ground truth, bullet symbols that were read as other symbols, field labels with underscores that were read as shortened words, and a few misread characters.

Table 7. NLP Skill Extraction (Recall) Results by Scan-Quality Condition

Component	Sample Size	Correctly Extracted/Classified	Result
NLP Skill Extraction (Recall): Baseline	191 expected skills across the 11 internal CV documents	159 skills correctly extracted	83.2%
NLP Skill Extraction (Recall): Easy scans	191 expected skills across the 11 internal CV documents	161 skills correctly extracted	84.3%
NLP Skill Extraction (Recall): Medium scans	191 expected skills across the 11 internal CV documents	156 skills correctly extracted	81.7%
NLP Skill Extraction (Recall): Hard scans	191 expected skills across the	104 skills correctly extracted	54.5%

	11 internal CV documents		
NLP Skill Extraction (Recall): Total (all conditions)	764 expected skills across the 44 scans	580 skills correctly extracted	75.9%

As presented in Table 7, the NLP skill extraction obtained a recall of 83.2% (159 out of 191 expected skills) on the baseline, 84.3% (161 out of 191) on the easy scans, 81.7% (156 out of 191) on the medium scans and 54.5% (104 out of 191) on the hard scans, or 75.9% (580 out of 764) across all conditions. Under the looser matching rule, the recall was 85.3%, 86.4%, 83.8% and 57.1%, respectively. From the baseline to the medium scans, the word-level accuracy decreased by 5.5 percentage points and the recall by 1.5 percentage points, while from the baseline to the hard scans, the recall decreased by 28.7 percentage points. Several skills were not extracted in any of the conditions, including Microsoft PowerPoint, AutoCAD 2D and 3D, Electrical Enclosures, Junction Boxes and Cable Glands.

Table 8. ML Skill Classification Results

Component	Sample Size	Correctly Extracted/Classified	Result
ML Skill Classification (TF-IDF + Logistic Regression)	106 human-verified labeled phrases	70 correctly classified (TP = 44, TN = 26, FP = 27, FN = 9)	66.0% (Precision = 62.0%, Recall = 83.0%, F1-score = 71.0%)

As presented in Table 8, the ML-based skill classification component, evaluated on 106 human-verified labeled phrases, achieved an accuracy of 66.0%, with a Precision of 62.0%, a Recall of 83.0% and an F1-score of 71.0%.

Table 9. Resource Recommendation Accuracy Results

Test Project	Employees Recommended	Employees Verified to Meet Requirements	Result
Industrial Facility Lighting Design and Documentation Project	5	5	100.0%
Industrial Electrical Engineering Drawings and	5	5	100.0%

Documentation Project			
Data Center UPS Installation	5	1	20.0%
UPS System Installation and Commissioning Support Project	5	2	40.0%
Industrial Electrical Proposal and Bid Preparation Project	5	0	0.0%
Solar Farm Electrical Integration	5	5	100.0%
Factory Electrical Maintenance Program	5	0	0.0%
Marine Electrical Equipment Supply	5	2	40.0%
Industrial Power Distribution Upgrade	5	1	20.0%
Industrial Lighting Design and Engineering Documentation Project	5	5	100.0%

As presented in Table 9, the recommendation engine is tested against ten sample projects, with top 5 ranked candidates evaluated per project against each project's requirements. The overall fill-rate across the 50 forced top 5 slots was 52.0% (26/50). When measured instead as Precision@K on the ranked recommendation list, the engine achieved a Precision@1 of 80.0% (8/10), a Precision@3 of 60.0% and a Precision@5 of 52.0% (26/50), the same figure as the overall fill-rate. The engine's own Recommended/Not Recommended classification for each candidate was compared against the human verified

project requirements across 50 candidate project pairs, result of a confusion matrix of TP = 26, TN = 24, FP = 0 and FN = 0 for a classification Accuracy, Precision and Recall of 100.0% each.

Project Testing Results

This section is present the results that are gather from the evaluation of the Automated Skill Extraction and Workload-Based Resource Recommendation System base on the ISO/IEC 25010 software quality standard. This is cover all seven (7) quality characteristics that are ask in the seventh research question of the study, namely Functional Suitability, Reliability, Usability, Performance Efficiency, Maintainability, Portability, and Security. These results is answer by the seventeen (17) respondents that are compose of thirteen (13) end users, three (3) technical experts and one (1) domain expert.

As discussed in the Data Collection Methods of Chapter 2, not all respondents is answer all seven (7) quality characteristics of the ISO/IEC 25010 model. The Functional Suitability, Reliability, and Usability items is answer by all seventeen (17) respondents, which is the end users, the domain expert, and the technical experts, since these three (3) characteristics is what the respondents themselves experience while using the system. The Performance Efficiency, Maintainability, Portability, and Security items, meanwhile, is only answer by the three (3) technical experts, since these four (4) characteristics is more of a technical and code-level concern that only the IT/technical experts are qualified

to judge (Lagman et al., 2025). The following tables and figures present the result that are gather from this evaluation.

Table 10. Demographic Profile of the Respondents

Respondent Group	Number of Respondents	Role/Designation	Relevant Experience
Domain Expert	1	Human Resources (HR)	7 years or more
Technical Expert	3	IT Expert	1–3 years to 4–6 years
End User	13	Employee (7), Human Resources (4), Resource Manager (1), Project Manager (1)	Less than 1 year to 7 years or more

As show in Table 8, one (1) domain expert, three (3) technical experts, and thirteen (13) end users take part in the evaluation. The end users are compose of seven (7) Employees, four (4) Human Resources personnel, one (1) Resource Manager, and one (1) Project Manager, with a range of experience from less than one (1) year to seven (7) years or more in their role. The technical experts, who are all IT Expert, have a range of experience from one (1) to three (3) years up to four (4) to six (6) years, while the domain expert, who is from

Human Resources (HR), have a relevant experience of seven (7) years or more in their role.

Table 11. *Summary of Functional Suitability Evaluation Results (n = 17)*

Indicator	Mean Score	Verbal Interpretation
Extracts and sorts skills correctly from the uploaded resume (CV).	4.53	Strongly Agree
Matches employee skills and past performance to what the project needs.	4.65	Strongly Agree
Shows correctly which employees are free and not yet assigned to a project.	4.65	Strongly Agree
Shows the applicant's uploaded resume next to their typed-in details for comparison.	4.71	Strongly Agree
Gives recommendations that are relevant and useful for assigning employees.	4.71	Strongly Agree
Adjusts its recommendations based on an employee's past performance ratings.	4.82	Strongly Agree
Its dashboards and reports give useful insights on workload and skill gaps.	4.59	Strongly Agree

As present in Table 9, all seven (7) indicators of Functional Suitability obtain a mean score that fall under the Strongly Agree range, with an overall mean of 4.66. The indicator that get the highest mean score is the system's ability to adjust its recommendation base on an employee's past performance rating, with a mean of 4.82, while the indicator on the accurate extraction and categorization of skills from the uploaded resume obtain the lowest mean among the seven indicators, with a mean of 4.53, which is still fall under the Strongly Agree range.

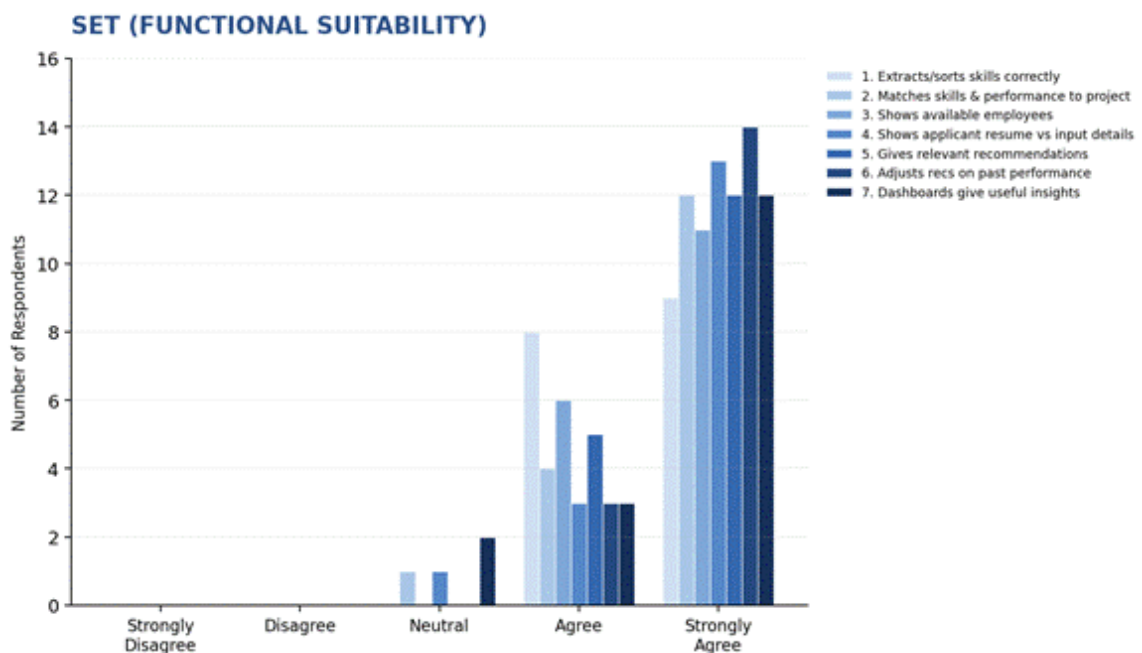


Figure 12. System Evaluation Tool (Functional Suitability)

Table 12. *Summary of Reliability Evaluation Results (n = 17)*

Indicator	Mean Score	Verbal Interpretation
Works consistently with no errors during normal use.	4.59	Strongly Agree
Recovers properly, with no data lost, when an error happens.	4.41	Agree
Gives the same skill extraction and recommendation results every time it is used with the same data.	4.71	Strongly Agree

As present in Table 10, the Reliability of the system obtain an overall mean of 4.57, which is fall under the Strongly Agree range. Among the three (3) indicators, the system's ability to recover properly and avoid data loss when an error occur obtain the lowest mean of 4.41, which is still fall under the Agree range, while the consistency of the skill extraction and recommendation results across repeated use obtain the highest mean of 4.71.

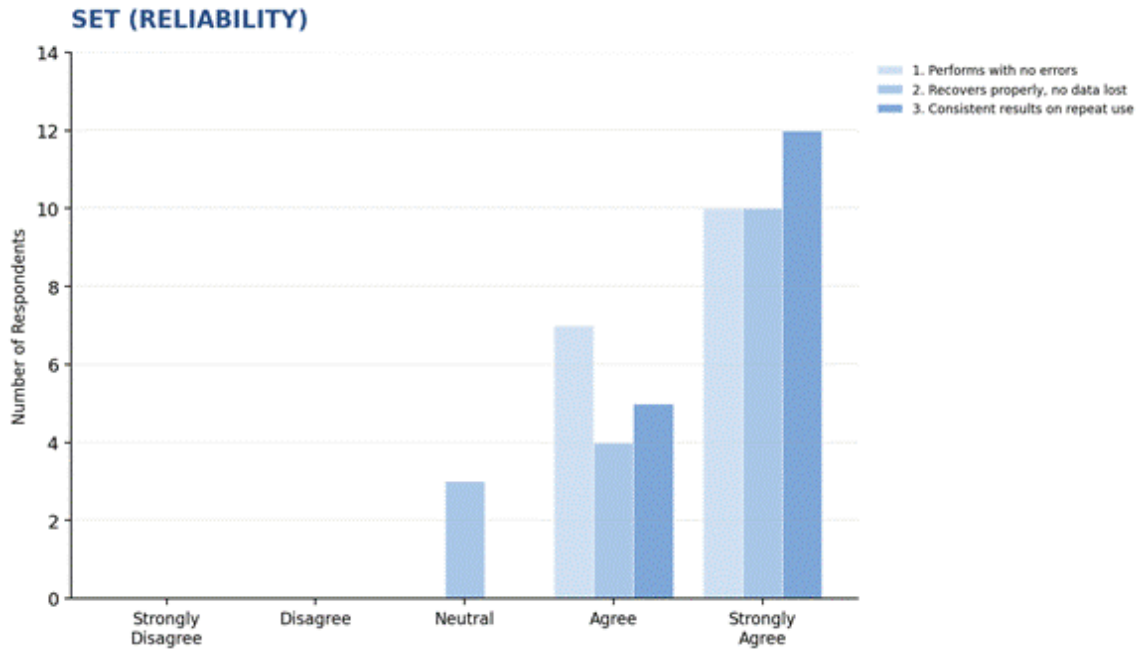


Figure 13. System Evaluation Tool (Reliability)

Table 13. Summary of Usability Evaluation Results (n = 17)

Indicator	Mean Score	Verbal Interpretation
Its interface is easy to navigate and understand for my role.	4.59	Strongly Agree
Lets me finish my task (e.g., uploading a resume, checking a recommendation) without needing help from others.	4.71	Strongly Agree
Gives a clear explanation on how a recommendation was made.	4.76	Strongly Agree

As present in Table 11, the Usability of the system obtain an overall mean of 4.69, which is fall under the Strongly Agree range. The indicator on the system giving a clear explanation on how a recommendation is generated obtain the highest mean of 4.76, while the ease of navigating and understanding the interface for the respondent's specific role obtain a mean of 4.59, the lowest among the three (3) indicators but is still fall under the Strongly Agree range.

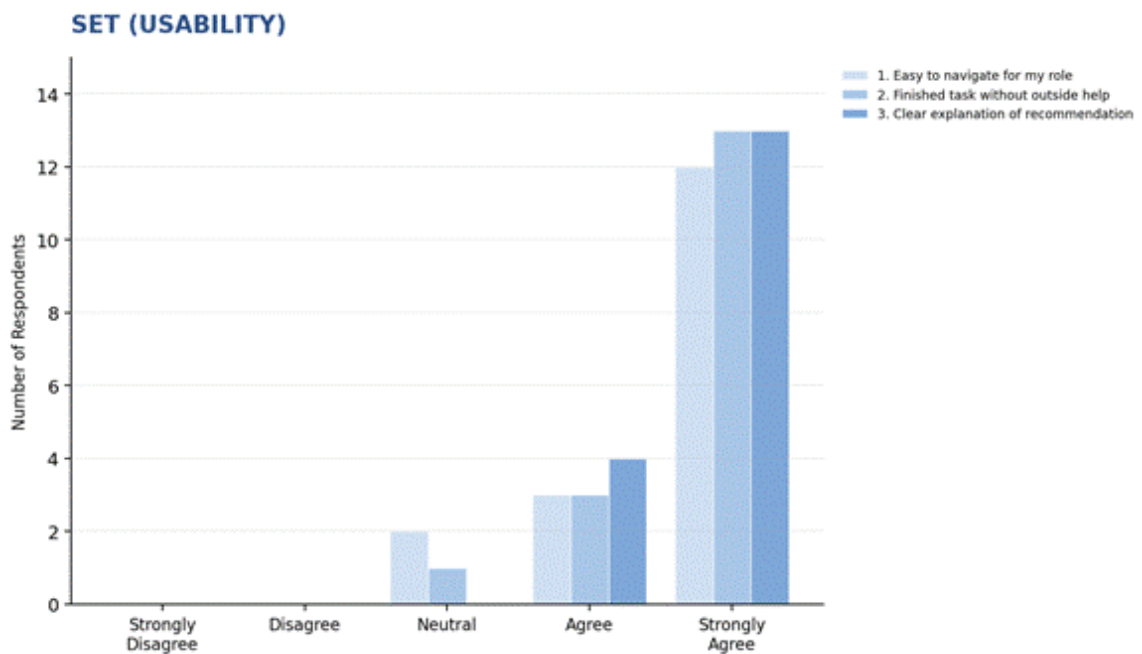


Figure 14. System Evaluation Tool (Usability)

Table 14. *Summary of Performance Efficiency Evaluation Results (n = 3, Technical Experts only)*

Indicator	Mean Score	Verbal Interpretation
Processes and returns the OCR/NLP skill extraction results within an acceptable amount of time.	4.67	Strongly Agree
Generates the ranked recommendation list without noticeable delay.	5.00	Strongly Agree
Dashboards and reports load and update in a reasonable amount of time.	5.00	Strongly Agree

As present in Table 12, the Performance Efficiency of the system obtain an overall mean of 4.89, which is fall under the Strongly Agree range. This is answer only by the three (3) technical experts, since this characteristic is about how the system perform on a technical level, which the end users do not directly experience or have the background to judge. Both the speed of the ranked recommendation list and the loading of the dashboards and reports obtain a perfect mean of 5.00, while the speed of the OCR/NLP skill extraction result obtain a mean of 4.67, still fall under the Strongly Agree range.

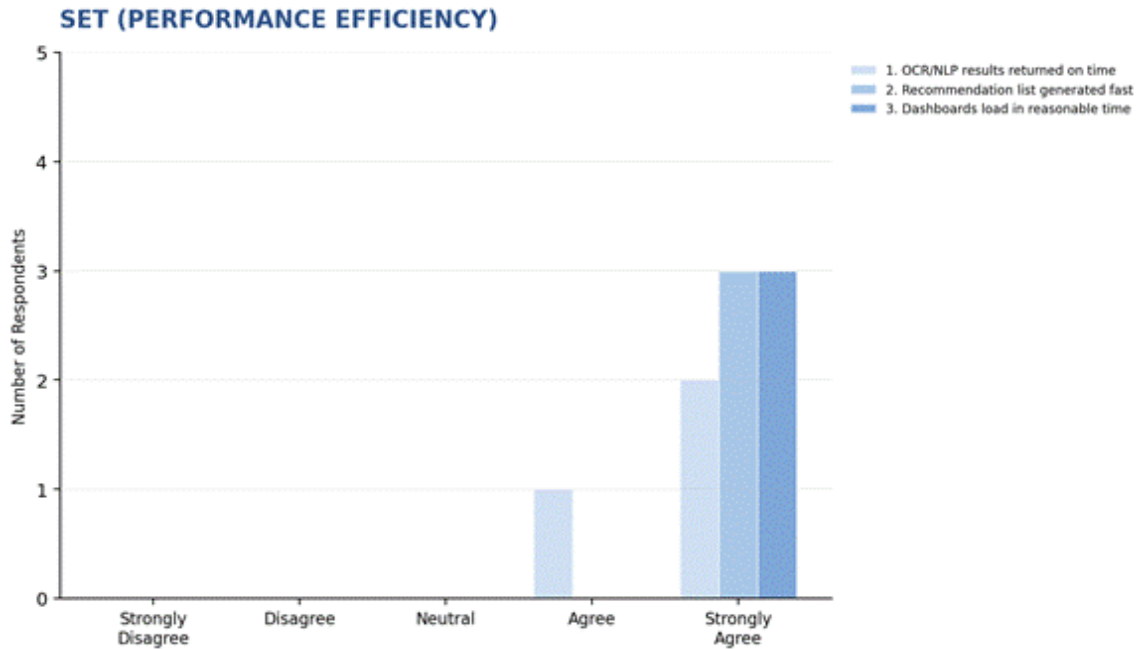


Figure 15. System Evaluation Tool (Performance Efficiency)

Table 15. Summary of Maintainability Evaluation Results ($n = 3$, Technical Experts only)

Indicator	Mean Score	Verbal Interpretation
Errors or issues in the system's modules (e.g., OCR, NLP, recommendation engine) can be easily found and diagnosed.	4.67	Strongly Agree
Individual parts of the system can be tested or updated without affecting the whole system.	4.33	Agree

As present in Table 13, the Maintainability of the system obtain an overall mean of 4.50, which is fall under the Agree range. This characteristic is also only answer by the three (3) technical experts, since it is about the system's source code and internal structure, which is a technical concern that an ordinary end user do not have access to. The ease of identifying and diagnosing errors in the system's modules obtain a mean of 4.67, while the ability to test or update individual parts of the system without affecting the whole obtain a mean of 4.33.

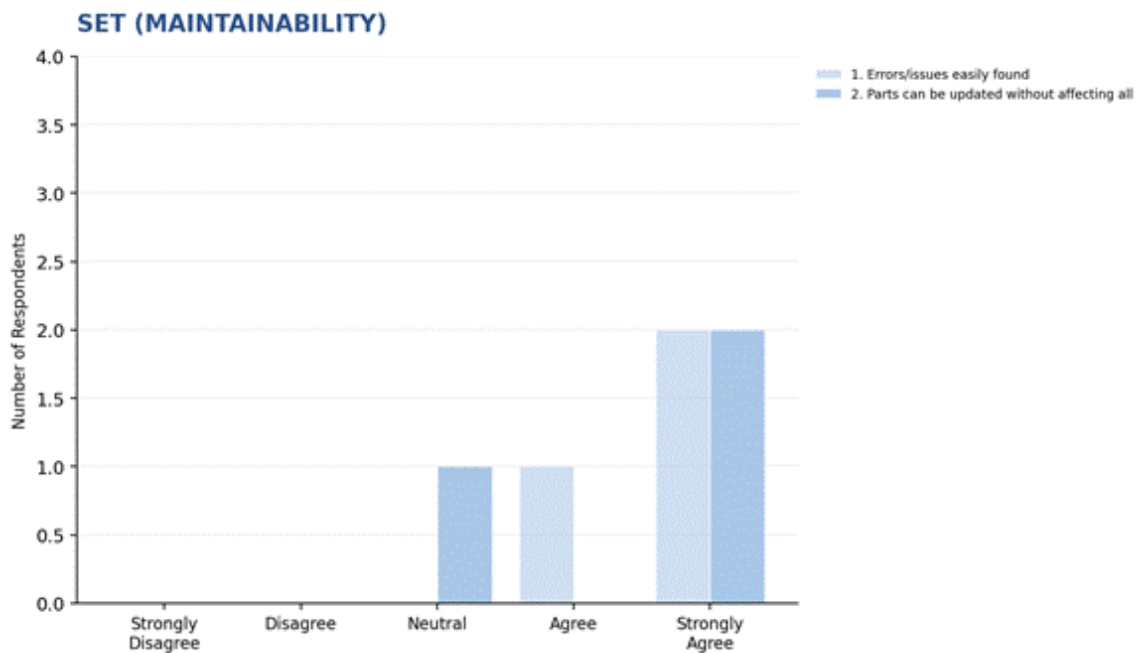


Figure 16. System Evaluation Tool (Maintainability)

Table 16. *Summary of Portability Evaluation Results (n = 3, Technical Experts only)*

Indicator	Mean Score	Verbal Interpretation
Is accessible and function properly through a standard web browser without needing special installation.	4.67	Strongly Agree
Its core functions still work as expected across different devices or browsers used for access.	4.33	Agree

As present in Table 14, the Portability of the system obtain an overall mean of 4.50, which is fall under the Agree range. This characteristic is likewise only answer by the three (3) technical experts, since it require testing the system across different devices, browsers, or environments, which is beyond what a typical end user would do. The system's accessibility through a standard web browser without special installation obtain a mean of 4.67, while its consistency across different devices or browsers obtain a mean of 4.33.

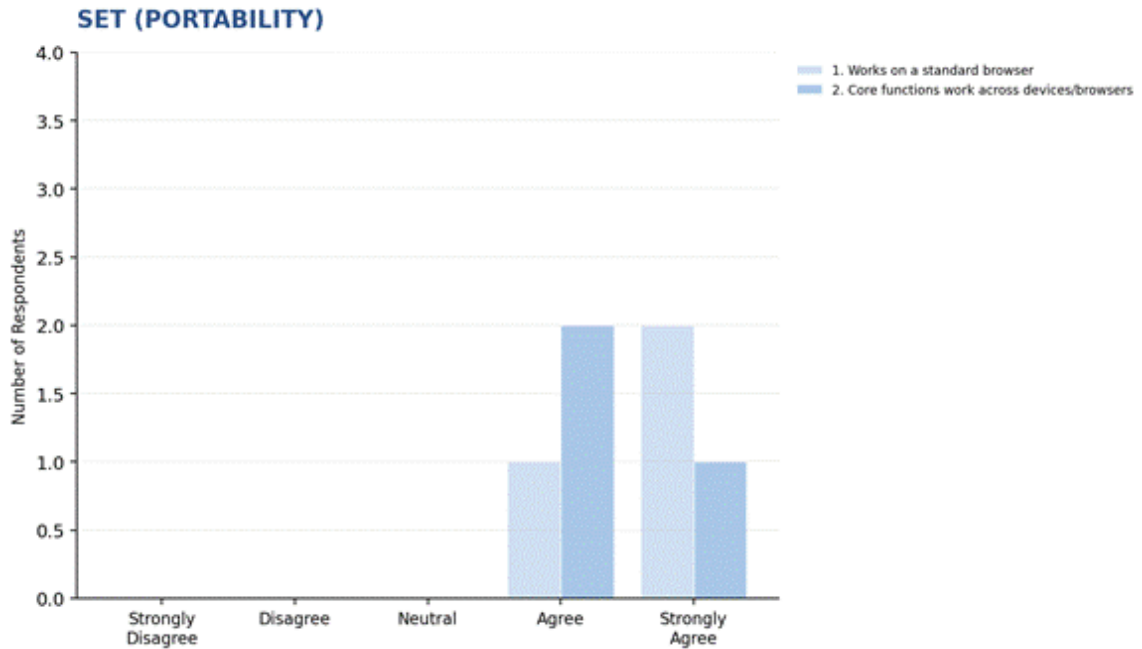


Figure 17. System Evaluation Tool (Portability)

Table 17. Summary of Security Evaluation Results ($n = 3$, Technical Experts only)

Indicator	Mean Score	Verbal Interpretation
Properly restricts access based on user role (Admin, Resource Manager, Project Manager, HR, Employee).	5.00	Strongly Agree
Its login and account protection features (e.g., lockout after failed attempts, session timeout) work as intended.	4.67	Strongly Agree

As present in Table 15, the Security of the system obtain an overall mean of 4.83, which is fall under the Strongly Agree range. This characteristic is answer only by the three (3) technical experts, since judging whether the system's role-based access control and login protection are properly and securely implemented require a technical background that an ordinary end user do not have. The system's restriction of access base on user role obtain a perfect mean of 5.00, while its login and account protection features obtain a mean of 4.67.

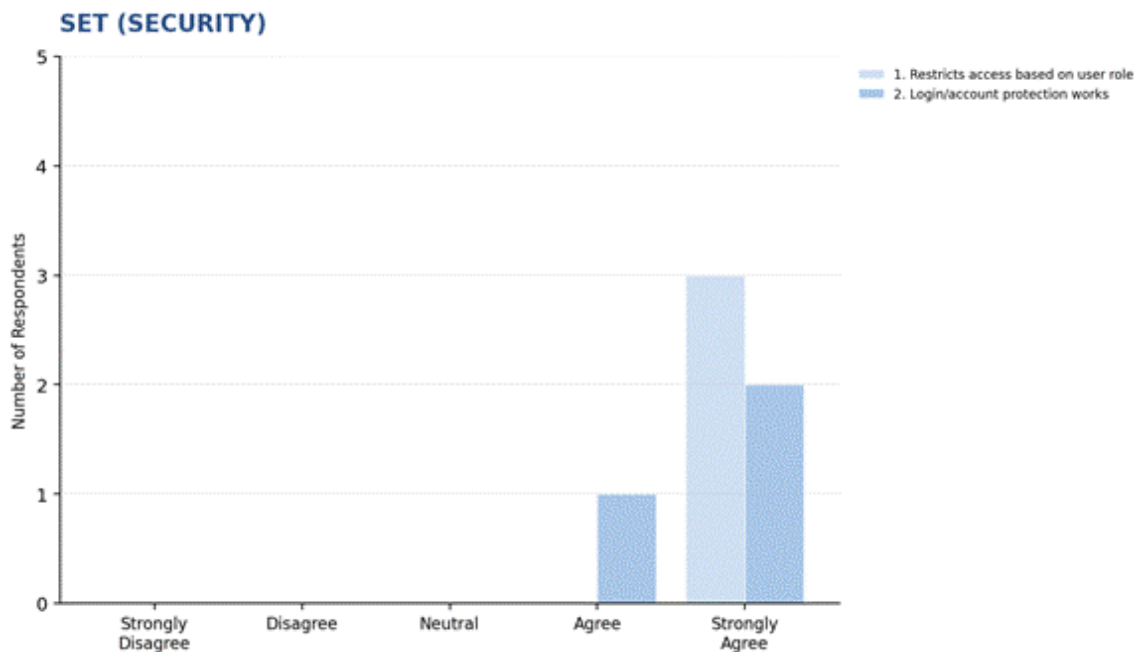


Figure 18. System Evaluation Tool (Security)

Table 18. Consolidated ISO/IEC 25010 Evaluation Summary

Quality Characteristic	Overall Mean Score	Verbal Interpretation
Functional Suitability	4.66	Strongly Agree
Reliability	4.57	Strongly Agree
Usability	4.69	Strongly Agree
Performance Efficiency	4.89	Strongly Agree
Maintainability	4.50	Agree
Portability	4.50	Agree
Security	4.83	Strongly Agree
Overall	4.66	Strongly Agree

Table 16 present the consolidated result of the ISO/IEC 25010 based evaluation, covering all seven (7) quality characteristics that answer the seventh research question of the study, namely Functional Suitability, Reliability, Usability, Performance Efficiency, Maintainability, Portability, and Security. The overall mean score across the seven (7) quality characteristics is 4.66, which is fall under the Strongly Agree range.

Aside from the quantitative ratings, feedback and observations that were gathered from the respondents during their interaction with the system are also consider as part of the results of this study.

Aside from the numerical ratings, several respondents also share their observations and suggestions through the open-ended question of the questionnaire. A Resource Manager respondent state that the system "was easy to navigate" but "may need some improvement." A Human Resources respondent share that they "can able to navigate the system without IT prior knowledge," while another Human Resources respondent describe the system as "highly functional and user friendly." A Technical Expert respondent suggest that "a chat assistant or help page could be really helpful for users, especially when they encounter issues or are confused when using it for the first time." Another Technical Expert respondent note that the system "works as intended" and is "designed around the company's specific needs, but also recommend "clearer error handling" and "another round of UI/UX cleanup," specifically pointing out that "some forms have low-opacity backgrounds, which allows the text behind them to show through while users are filling them out."

CHAPTER 4

CONCLUSIONS AND RECOMMENDATIONS

Introduction

This chapter presents the conclusions that are drawn from the results that are gathered in Chapter 3, and the recommendations that are formulated based on these conclusions. The conclusions are arranged according to the order of the Statement of the Problem that is present in Chapter 1, while the recommendations are divided into recommendations for action, recommendations for future research, and recommendations for system or process improvement.

Conclusion

The main purpose of this study is to develop an automated skill extraction and workload-based resource recommendation system for WEA to improve workforce allocation, task matching and operational efficiency. The system is evaluated by seventeen (17) purposively selected respondents, which is composed of thirteen (13) end users, three (3) technical experts and one (1) domain expert, using the ISO/IEC 25010-based Likert-scale questionnaire, and it is also evaluated through the functional testing of five hundred seven (507) test cases and the accuracy testing of the OCR, NLP, skill classification and recommendation engine. Based on these results, the following conclusions are drawn in the order of the Statement of the Problem.

For the first problem, on what system can be develop for WEA to automate skill extraction and support intelligent resource allocation, the study develop a web-based Skill-Extraction and Workload-Based Resource Recommendation System that integrate OCR, NLP and a weighted recommendation engine, together with the Applicant, Employee, Human Resources, Project Manager, Resource Manager, Admin and Super Admin modules. In the functional testing, four hundred ninety-five (495) out of five hundred seven (507) test cases is pass, with an overall pass rate of 97.63%, and the respondents rate the Functional Suitability of the system with an overall mean of 4.66, which is Strongly Agree.

For the second problem, on how to extract and classify employee skills from internal resume documents via OCR and NLP, the system use Tesseract OCR to convert the internal CV into text, and a NLP pipeline with a TF-IDF and Logistic Regression classifier to identify and classify the skills. On the eleven (11) internal CV documents, evaluated under a baseline and three simulated scan-quality levels, the OCR component achieve a word-level accuracy of 77.3% on the baseline, 77.9% on the easy scans, 71.8% on the medium scans and 56.8% on the hard scans (Word Error Rate of 30.3%, 30.4%, 35.8% and 60.9%, respectively), the NLP skill extraction correctly identify 83.2%, 84.3%, 81.7% and 54.5% of the 191 expected skills under the same conditions, and the ML skill classification, that is evaluate on 106 human-verified labeled phrases, achieve an accuracy of 66.0%, a Precision of 62.0%, a Recall of 83.0% and an F1-score of

71.0%. The results also show that the errors are not caused by the scan quality alone, since the baseline without degradation obtained a word-level accuracy of 77.3%, and that the skill extraction remained above 80% until the scans were heavily degraded, so the poor-quality uploads should be reviewed by the Employee. The respondents also rate the indicator on the accurate extraction and sorting of skills from the uploaded resume with a mean of 4.53, which is Strongly Agree.

For the third problem, on how to analyze workload distribution across employees to identify underutilization and overloading, the system compute the Workload Score of each employee base on the priority weight of the active tasks, and convert it into an Availability Factor using a sigmoid function. The respondents rate the indicator on the dashboards and reports that give useful insights on workload and skill gaps with a mean of 4.59, and the indicator on the correct showing of the employees that are free and not yet assigned to a project with a mean of 4.65, both is Strongly Agree, while the Resource Manager Module pass all fifty (50) of its test cases.

For the fourth problem, on how to generate accurate and fair resource recommendations based on skill matching, availability and workload conditions, the Recommendation Engine combine the Matching Score, the Availability Factor and the Historical Performance Factor using a weighted sum of 60%, 25% and 15%, respectively, to rank the candidates. In the accuracy testing on ten (10)

sample projects, the engine obtain a Precision@1 of 80.0%, a Precision@3 of 60.0% and a Precision@5 of 52.0%, which is the same as the overall fill-rate of twenty-six (26) out of fifty (50) top five slots, and its Recommended/Not Recommended classification obtain an Accuracy, Precision and Recall of 100.0% each. The respondents also rate the indicator on the matching of the employee skills and past performance to the project needs with a mean of 4.65, and the indicator on the relevance and usefulness of the recommendations with a mean of 4.71, both is Strongly Agree.

[PLACEHOLDER — TEXT: The fairness part of the fourth problem is not yet answerable, since the System Evaluation Tool and the testing in Chapter 3 have no item or measure for the fairness of the recommendations (for example, how evenly the assignments are distribute among the employees). Insert the fairness result here once it is measure, or revise the wording of the fourth problem in Chapter 1.]

For the fifth problem, on how to continuously learn and adapt from historical assignments and performance outcomes to improve recommendation accuracy, the system collect the Client's Performance Rating after every completed project and use it as the Historical Performance Factor in the next recommendation of the same employee, and it also retrain the skill classifier once twenty (20) human-verified examples is accumulate. The Client Feedback Module pass all eight (8) of its test cases, and the respondents rate the indicator on the adjustment of the recommendations base on the employee's past performance rating with a mean of 4.82, the highest among the Functional

Suitability indicators, and the indicator on the consistency of the results across repeated use with a mean of 4.71.

[PLACEHOLDER — TEXT: The improvement of the recommendation accuracy over time is not yet answerable, since Chapter 3 has no before-and-after result that show the accuracy is increase as the client ratings and human-verified skills accumulate. Insert this result here once it is measure.]

For the sixth problem, on how data analytics can be integrated to provide insights into workforce capacity, skill gaps and strategic planning, the Resource Manager module include dashboards and reports that display the department-level workload and utilization. The respondents rate the indicator on the usefulness of the dashboards and reports for identifying the workforce capacity and skill gaps with a mean of 4.59, which is Strongly Agree, the three (3) technical experts rate the loading of the dashboards and reports with a mean of 5.00, and the Resource Manager Module pass all fifty (50) of its test cases.

For the seventh problem, on the level of acceptability of the system base on the ISO/IEC 25010 software quality model, the system obtain an overall mean of 4.66, which is Strongly Agree. In terms of the seven (7) quality characteristics, the Functional Suitability obtain a mean of 4.66, the Reliability 4.57, the Usability 4.69, the Performance Efficiency 4.89, the Maintainability 4.50, the Portability 4.50 and the Security 4.83. The Functional Suitability, Reliability, Usability, Performance Efficiency and Security is Strongly Agree, while the Maintainability and Portability is Agree. The Performance Efficiency, Maintainability, Portability and Security is answer only by the three (3) technical experts.

Overall, the study show that the developed Skill-Extraction and Workload-Based Resource Recommendation System is acceptable to its respondents in all seven (7) quality characteristics of the ISO/IEC 25010, and it give WEA a working tool that extract the skills from the internal CV of the employees and rank the employees for a project assignment base on their skills, availability and past performance.

Recommendations

Base on the conclusions that are present above, the following recommendations is offer. Each recommendation is trace to the conclusion that is state in the previous section.

Recommendations for Action

Base on the conclusion for the first problem, the researchers recommend that the three (3) failed test cases and the nine (9) test cases that is not yet execute should be resolve and re-test before the system is use by WEA, so that all of the ten (10) modules are verify to perform as expected. Base on the conclusion for the second problem, WEA is recommend to keep its internal CV in the standardized format and to let the Employees review the skills that is flag by the system in the Skill Review screen, since the OCR word-level accuracy falls to 56.8% on the hard scans and the classifier Precision is 62.0%, and every skill that is approve or reject by the Employee is add as a human-verified example

that help the classifier to improve. WEA is also recommend to ask the Employees to upload a clear file, since the skill extraction recall decrease from 81.7% on the medium scans to 54.5% on the hard scans.

Base on the conclusion for the fourth problem, the Resource Manager is recommend to review the ranked list together with the skill match and availability of each candidate before the final assignment, since the Precision@5 of the recommendation is 52.0%. Base on the conclusion for the fifth problem, the Project Managers is recommend to make sure that the Client submit the Performance Rating after every completed project, so that the Historical Performance Factor is base on the actual rating of the employee instead of the neutral value of 0.70.

Recommendations for Future Research

Base on the conclusion for the fourth and fifth problem, future studies is recommend to measure the fairness of the recommendations, such as how evenly the assignments is distribute among the employees, and to measure if the recommendation accuracy improve over time as the client ratings and human-verified skills are accumulate, since these is not yet measure in this study. Base on the conclusion for the second problem, future studies may also expand the OCR and NLP to support other resume formats aside from the standardized internal CV of WEA, such as Word documents and non-standard layouts, and to evaluate the OCR and NLP using actual scanner and phone-camera captures, since the scans in this study were simulated from eleven (11) documents of a

single format. Base on the conclusion for the seventh problem, future studies is recommend to increase the number of respondents beyond seventeen (17), especially the technical experts, since the Performance Efficiency, Maintainability, Portability and Security is rate by only three (3) technical experts.

Recommendations for System or Process Improvement

Base on the conclusion for the second problem, the OCR and the skill classification of the system may be improve further to lower the Word Error Rate of 60.9% on the hard scans and to raise the 62.0% Precision, such as adding an image-quality check that ask the Employee to upload a clearer file before the OCR process, and adding more human-verified examples for the classifier. Base on the conclusion for the fourth problem, the skill matching of the Recommendation Engine may be review for the projects where only few or none of the top five candidates meet the requirements, such as the Industrial Electrical Proposal and Bid Preparation Project and the Factory Electrical Maintenance Program, where none of the five (5) recommended employees is verify to meet the requirements.

Base on the conclusion for the first problem, the duplicate application prevention in the Applicant Module, the required Role field in the Admin Module, and the confirmation message after saving the skill feedback in the Employee Module may be fix, since these is the three (3) test cases that fail. Base on the conclusion for the seventh problem, the suggestions of the respondents, such as adding a chat assistant or help page for first-time users, clearer error handling,

and a UI/UX cleanup of the forms with low-opacity backgrounds, may be consider, together with the improvement of the error recovery (mean of 4.41), the testing or update of the individual parts of the system (mean of 4.33), and the consistency across different devices and browsers (mean of 4.33), that is the lowest rated indicators.

REFERENCE LIST

- Alberto, L., Lebis, A., Karami, A., Vermeulen, M., Fleury, A., & Van De Wiele, C. (2025). A recommender system of postgraduate courses based on soft skills: A psychometric-inspired approach. *International Journal of Artificial Intelligence in Education*. <https://doi.org/10.1007/s40593-025-00463-z>
- Alsaif, S. A., Hidri, M. S., Ferjani, I., Eleraky, H. A., & Hidri, A. (2022). NLP-based bidirectional recommendation system: Towards recommending jobs to job seekers and resumes to recruiters. *Big Data and Cognitive Computing*, 6(4), 147. <https://doi.org/10.3390/bdcc6040147>
- Andrade, C. (2021). The inconvenient truth about convenience and purposive samples. *Indian Journal of Psychological Medicine*, 43(1), 86–88. <https://doi.org/10.1177/0253717620977000>
- Bai, B., Dai, H., Zhang, D., Zhang, F., & Hu, H. (2022). The impacts of algorithmic work assignment on fairness perceptions and productivity: Evidence from field experiments. *Manufacturing & Service Operations Management*, 24(6), 3060–3078. <https://doi.org/10.1287/msom.2022.1120>
- Banerjee, A., Banik, P., & Wörndl, W. (2023). A review on individual and multistakeholder fairness in tourism recommender systems. *Frontiers in Big Data*, 6, Article 1168692. <https://doi.org/10.3389/fdata.2023.1168692>

Barletta, V. S., Caivano, D., Colizzi, L., Dimauro, G., & Piattini, M. (2023). Clinical-chatbot AHP evaluation based on "quality in use" of ISO/IEC 25010. *International Journal of Medical Informatics*, 170, Article 104951. <https://doi.org/10.1016/j.ijmedinf.2022.104951>

Biol, I. J. A., & Abalorio, C. C. (2024). Utilizing Named Entity Recognition for Web-Based Resume Scoring. *International Journal of Engineering Trends and Technology*, 72(7), 381–387. <https://doi.org/10.14445/22315381/IJETT-V72I7P142>

Boukhari, S., Tlili, M., & Abdellatif, T. (2024). Skill matching at scale: Freelancer-project alignment for efficient multilingual candidate retrieval. arXiv. <https://arxiv.org/abs/2409.12097>

Brill, K. (2025, July 9). What is Rapid Application Development (RAD)? Built In. <https://builtin.com/articles/rapid-application-development>

Cacatian, L. B., Esguerra, J. E. C., Estacaan, A. G., Marasigan, J. B., Sinosa, M. A. B., & Tatco, J. A. V. (2025). Automated innovation: Exploring the integration of artificial intelligence (AI) in human resource recruitment in Metro Manila. ResearchGate. https://www.researchgate.net/publication/387787089_Automated_Innovation_Exploring_The_Integration_of_Artificial_Intelligence_AI_In_Human_Resource_Recruitment_in_Metro_Manila

Carnovalini, F. (2025). Popularity bias in recommender systems. *Information*, 16(2), 151. <https://doi.org/10.3390/info16020151>

Clavié, B., & Soulié, G. (2023). Large language models as batteries-included zero-shot ESCO skills matchers. *arXiv*. <https://arxiv.org/abs/2307.03539>

De Campos, L. M., Fernandez-Luna, J. M., & Huete, J. F. (2024). An explainable content-based approach for recommender systems: A case study in journal recommendation for paper submission. *User Modeling and User-Adapted Interaction*. <https://doi.org/10.1007/s11257-024-09400-6>

Decorte, J. J., Van Haute, J., Demeester, T., & Develder, C. (2024). SkillMatch: Evaluating self-supervised learning of skill relatedness. *arXiv*. <https://arxiv.org/abs/2410.05006>

Deshmukh, A., & Raut, A. (2024). Applying BERT-based NLP for automated resume screening and candidate ranking. *Annals of Data Science*. <https://doi.org/10.1007/s40745-024-00524-5>

Du, W., & Chen, X. (2025). A new method of task aggregation and optimization allocation for multiple groups collaborative task networks. *Scientific Reports*, 15, Article 27566. <https://doi.org/10.1038/s41598-025-11762-9>

Frazzetto, P., Ul Haq, M. U., Fabris, F., & Sperduti, A. (2025). From text to talent: A pipeline for extracting insights from candidate profiles. *arXiv*. <https://arxiv.org/abs/2503.17438>

Gavrilescu, M., Leon, F., & Minea, A. A. (2025). Techniques for transversal skill classification and relevant keyword extraction from job advertisements. *Information*, 16(2), 167. <https://doi.org/10.3390/info16020167>

Gan, C., Zhang, Q., & Mori, T. (2024). Application of LLM agents in recruitment: A novel framework for resume screening. *arXiv preprint arXiv:2401.08315*. <https://arxiv.org/abs/2401.08315>

Gómez, E., Contreras, D., Boratto, L., & Salamó, M. (2025). Enhancing recommender systems with provider fairness through preference distribution awareness. *International Journal of Information Management Data Insights*, 5(1), Article 100311. <https://doi.org/10.1016/j.jjime.2024.100311>

Gotz, G. (2023, December 29). An approach to implement user-based recommendation systems with small-sized data sets. ResearchGate. <https://www.researchgate.net/publication/376819602>

Hanif, A. A., Belhfandho, N., & Nurdiana, E. (2025). Analysis of the effectiveness of competency-based performance assessment systems in companies in Cirebon Regency. *Siber International Journal of Digital Business*, 3(1), 45–52. <https://doi.org/10.38035/sijdb.v3i1.273>

Helli, S. S., Tanberk, S., & Cavsak, S. N. (2023). Resume information extraction via post-OCR text processing. In 2023 IEEE 3rd Mysore Sub Section International Conference (MysuruCon) (pp. 1–6). IEEE. <https://doi.org/10.1109/MysuruCon59703.2023.10296715>

Huang, L., Gao, Y., & Xiao, F. (2026). Balancing workload fairness in task assignment: Modeling via piecewise linear approximation. *Applied Sciences*, 16(4), 1747. <https://doi.org/10.3390/app16041747>

Ilyas, M., & Ahmad, W. (2023). Evaluation of web-based information system quality using ISO/IEC 25010. *International Journal of Information System and Technology*, 7(2), 234–241. <https://ijistech.org/ijistech/index.php/ijistech/article/view/234>

Imran, M., & Bhutto, A. (2023). RAD-based web application development for small and medium enterprises. *International Journal of Scientific Research in Computer Science Engineering and Information Technology*, 9(3). <https://ijsrcseit.com/paper/CSEIT2390345.pdf>

ISO/IEC/IEEE. (2022). ISO/IEC/IEEE 26514:2022 — Systems and software engineering: Design and development of information for users. International Organization for Standardization. <https://www.iso.org/standard/79060.html>

Janani, M., Laya, H., & Mathishree, B. (2025). OCR text recognition and recommendation using machine learning. In 2025 International Conference on Visual Analytics and Data Visualization (ICVADV). IEEE. https://www.researchgate.net/publication/391038785_OCR_Text_Recognition_and_Recommendation_Using_Machine_Learning

Jemal, I., Naoussi Sijou, W. A., & Chikhaoui, B. (2024). Multi-modal recommender system for predicting project manager performance within a

competency-based framework. *Frontiers in Big Data*, 7, Article 1295009.
<https://doi.org/10.3389/fdata.2024.1295009>

Jin, D., Wang, L., Zhang, H., Zheng, Y., Ding, W., Xia, F., & Pan, S. (2023). A survey on fairness-aware recommender systems. arXiv.
<https://doi.org/10.48550/arXiv.2306.00403>

Kashif, M., & Kumar, K. R. P. (2024). NLP-powered resume matching for recruitment. *International Journal of Advanced Research in Computer and Communication Engineering*, 13(9), 106–113.
<https://www.ijfmr.com/papers/2024/6/31742.pdf>

Kasri, M. A., Muharom, L. A., & Wardani, R. C. (2024, April 12). Evaluation of the new student candidates admission information system using ISO/IEC 25010 model. OSF Preprints. <https://doi.org/10.31219/osf.io/e8znx>

Kishore, C. R., & Sreerala, T. (2025). AI-powered resume analyzer and job matching system. *International Journal of Research in Intelligent Systems*, 10(11), 382–387. <https://doi.org/10.51584/IJRIAS>

Lagman, A. C., Ramirez, I. C. R., Esteban, A. P., Rivera, R. P. L., Santos, R. D., & Juliano, J. J. (2025). Development and evaluation of an enterprise-level information system for digital governance in Philippine SUCs using agile software methodology and ISO/IEC 25010 software quality model. *2024 IEEE 16th International Conference on Humanoid, Nanotechnology, Information*

Technology, Communication and Control, Environment, and Management (HNICEM), 1–6. <https://doi.org/10.1109/hnicem64917.2024.11258693>

Li, Y., Chen, H., Xu, S., Ge, Y., Tan, J., Liu, S., & Zhang, Y. (2022). Fairness in recommendation: A survey. *arXiv*. <https://doi.org/10.48550/arXiv.2205.13619>

Lima, S. A., & Rahman, M. M. (2024). Algorithmic fairness in HRM: Balancing AI-driven decision making with inclusive workforce practices. *Journal of Information Systems Engineering and Management*, 9(4s). <https://doi.org/10.52783/jisem.v9i4s.13251>

Magron, A., Dai, A., Zhang, M., Montariol, S., & Bosselut, A. (2024). JobSkape: A framework for generating synthetic job postings to enhance skill matching. In E. Hruschka, T. Lake, N. Otani, & T. Mitchell (Eds.), *Proceedings of the first workshop on natural language processing for human resources (NLP4HR 2024)* (pp. 43–58). Association for Computational Linguistics. <https://aclanthology.org/2024.nlp4hr-1.4/>

Malashin, I., Masich, I., Tynchenko, V., Gantimurov, A., Nelyub, V., & Borodulin, A. (2024). Image text extraction and natural language processing of unstructured data from medical reports. *Machine Learning and Knowledge Extraction*, 6(2), 1361–1377. <https://doi.org/10.3390/make6020064>

Meyliana, A., Rapiyanta, P. T., & Andriani, A. (2024). Application of the Rapid Application Development (RAD) method for web-based financial management

and wood inventory using Codelgniter. *ARRUS Journal of Engineering and Technology*, 4(1), 81–89. <https://doi.org/10.35877/jetech2722>

Mueller, V., & Schaaff, K. (2025). AI-based recommender system for employee-project matching of IT specialists. In T. Schlippe, E. C. K. Cheng, & T. Wang (Eds.), *Artificial intelligence in education technologies: New development and innovative practices* (pp. 296–311). Springer. https://doi.org/10.1007/978-981-97-9255-9_20

Najam, R., & Faizullah, S. (2023). Analysis of recent deep learning techniques for Arabic handwritten-text OCR and post-OCR correction. *Applied Sciences*, 13(13), 7568. <https://doi.org/10.3390/app13137568>

Naveed, Z., Nisar, B., Saifullah, D. M., & Iqbal Baig, J. (2024). Resume ranking using natural language processing. *Journal of Computers and Intelligent Systems*, 2(1), 61–70. <https://dergipark.org.tr/en/download/article-file/4159614>

Nidhra, S., & Dondeti, J. (2012). Black box and white box testing techniques – A literature review. *International Journal of Embedded Systems and Applications*, 2(2), 29–48. <https://doi.org/10.5121/ijesa.2012.2204>

Ningsih, R., Azhar, Y., & Munarko, Y. (2024). Rekomendasi lowongan pekerjaan dari portal bursa lowongan kerja memanfaatkan cosine similarity dan simple additive weighting. *Jurnal Repositor*, 2(5). <https://doi.org/10.22219/repositor.v2i5.30635>

Novita, D., Rapiyanta, P. T., & Andriani, A. (2023). Implementation of Rapid Application Development in Android-based school information system. Raden Intan State Islamic University. <https://ejournal.radenintan.ac.id>

Nugraha, U., & Rachman, M. A. N. (2024, December 29). Quality evaluation study. In Proceedings of the Widyatama International Conference on Engineering 2024 (WICOENG 2024). Atlantis Press. <https://www.atlantis-press.com/proceedings/wicoeng-24/126007237>

Nuzula, M. I. F., & Rochimah, S. (2023). Evaluation of service quality in human resource information systems using the ISO/IEC 25010. In 2023 International Seminar on Application for Technology of Information and Communication (iSemantic) (pp. 219–224). IEEE. <https://doi.org/10.1109/iSemantic59612.2023.10295365>

Parlika, R., Afifudin, M., Pradana, I. A., Wiratama, Y. D. W., & Holis, M. N. (2023). Literature study on the efficiency of the Rapid Application Development model in software development. POSITIF: Jurnal Sistem dan Teknologi Informasi, 8(2), 64–73. <https://doi.org/10.31961/positif.v8i2.1329>

Pinto-DelaCadena, P. A., Liern, V., & Vinueza-Cabezas, A. (2024). A comparative analysis of multi-criteria decision methods for personnel selection: A practical approach. Mathematics, 12(2), 324. <https://doi.org/10.3390/math12020324>

Poncio, F. P. (2024). Navigating techniques in job recommender systems on internship profile matching: A systematic review. *Journal of Research in*

Innovative Teaching & Learning, 17(2), 352–367.
<https://doi.org/10.1108/JRIT-01-2024-0016>

R, G. N., Vimal, M., Akkim, M., Tharnesh, A. R., & Vigneshkumar, P. (2026). AI resume analyzer and job recommendation system with NLP integration. *International Journal of Engineering Development and Research*, 14(1), 463–470. <https://rjwave.org/ijedr/papers/IJEDR26A1116.pdf>

Rampisela, T. V., Maistro, M., Ruotsalo, T., & Lioma, C. (2023). Evaluation measures of individual item fairness for recommender systems: A critical study. *arXiv*. <https://doi.org/10.48550/arXiv.2311.01013>

Riadi, I., Yudhana, A., & Elvina, A. (2024). Analysis impact of Rapid Application Development method on development cycle and user satisfaction: A case study on web-based registration service. *Scientific Journal of Informatics*, 11(1). <https://journal.unnes.ac.id/nju/sji/article/view/49590>

Rojas, H., Renteria, R., Duran, V. M., Gutiérrez, Y. T., Cabrera, M. J. I., & Aminuddin, A. (2025). Mapping the evolution and future directions of ISO/IEC 25010: A bibliometric and thematic analysis. *Engineering, Technology & Applied Science Research*, 15(5), 27530–27541. <https://doi.org/10.48084/etasr.11772>

Roy, D., & Dutta, M. (2022). A systematic review and research perspective on recommender systems. *Journal of Big Data*, 9(1), 59. <https://doi.org/10.1186/s40537-022-00592-5>

Saatçı, M., Kaya, R., & Ünlü, R. (2024). Resume screening with natural language processing (NLP). *Alphanumeric Journal*, 12(2), 121–140. <https://doi.org/10.17093/alphanumeric.1536577>

Shaker, A. A., Austin, S. F., Jørgensen, M. S., Sørensen, J. A., Bechmann, H., Kinnerup, H. E., Petersen, C. J., Olsen, R. K., & Simonsen, E. (2023). Facilitating in-house mobile app development within psychiatric outpatient services for patients diagnosed with Borderline Personality Disorder: Rapid application development approach. *JMIR Human Factors*, 10, Article e46928. <https://doi.org/10.2196/46928>

Sohail, M. A., Masood, S., & Iqbal, H. (2024). Deciphering the underserved: Benchmarking LLM OCR for low-resource scripts. *arXiv*. <https://doi.org/10.48550/arXiv.2412.16119>

Stankovic, J. (2024). Rapid Application Development in modern software engineering. *ResearchGate*. <https://www.researchgate.net/publication/379548372>

Sujon, K. M., Hassan, R., Choi, K., & Samad, M. A. (2025). Accuracy, precision, recall, F1-score, or MCC? Empirical evidence from advanced statistics, ML, and XAI for evaluating business predictive models. *Journal of Big Data*, 12, Article 268. <https://doi.org/10.1186/s40537-025-01313-4>

Susanto, A., & Andriana, D. (2023). Quality measurement of academic information system based on ISO/IEC 25010. *Journal of Informatics and Applied*

Mathematics, 6(1).

<https://journal.stmikjayakarta.ac.id/index.php/jisamar/article/view/819>

Tang, F., Zhu, R., Yao, F., Wang, J., Luo, L., & Li, B. (2025). Explainable person–job recommendations: Challenges, approaches, and comparative analysis. *Frontiers in Artificial Intelligence*, 8, Article 1660548. <https://doi.org/10.3389/frai.2025.1660548>

Thangam, S., Kumaran, U., Biswas, D., Sneha, B., Nadipalli, S., & Raja, S. (2023). Text post-processing on optical character recognition output using natural language processing methods. In *2023 IEEE 3rd Mysore Sub Section International Conference (MysuruCon)* (pp. 1–6). IEEE. <https://doi.org/10.1109/MysuruCon59703.2023.10296715>

Varshith Reddy, K. S., Kiran, N. J., Likhith Gowda, T. R., Amanullah, M., & Harini, S. (2025). Resume analyzer and job recommendation system. *Iconic Research and Engineering Journals*, 8(9), 282–287. <https://www.irejournals.com/paper-details/1707439>

Wang, L., Mistry, S., Hasan, A. A., Hassan, A. O., Islam, Y., & Osei, F. A. J. (2023). Implementation of a collaborative recommendation system based on multi-clustering. *Mathematics*, 11(6), 1346. <https://doi.org/10.3390/math11061346>

Wang, Y. (2025). Improving user-oriented fairness in recommendation systems. *Journal of Systems and Software*, 223, Article 112387. <https://doi.org/10.1016/j.jss.2025.112387>

Wei, S., Tong, Y., Zhou, Z., Liu, Q., Zhang, L., Zeng, Y., & Ye, J. (2023). Towards capacity-aware broker matching: From recommendation to assignment. arXiv. <https://arxiv.org/abs/2303.03024>

Yang, T. (2025). A novel framework for TER allocation using multilayer perceptron and intuitionistic fuzzy Z numbers for talent management. *Scientific Reports*, 15, Article 15055. <https://doi.org/10.1038/s41598-025-15055-z>

Youssri, A., Amin, E. A., Gheith, M., & Khater, A.-S. (2024). The future ERP systems: Improve employee performance evaluation using machine learning. In A. E. Hassanien, A. Darwish, M. F. Tolba, & V. Snasel (Eds.), *Proceedings of the 10th International Conference on Advanced Intelligent Systems and Informatics 2024 (AISI 2024)* (pp. 13–21). Springer. https://doi.org/10.1007/978-3-031-71619-5_2

Zhao, Y., Wang, Y., Liu, Y., Cheng, X., Aggarwal, C. C., & Derr, T. (2025). Fairness and diversity in recommender systems: A survey. *ACM Transactions on Intelligent Systems and Technology*, 16(1), Article 2, 1–28. <https://doi.org/10.1145/3664928>

Zhou, X., Zhang, L., Zhang, H., Zhang, Y., Zhang, X., Zhang, J., & Shen, Z. (2024). Advancing sustainability via recommender systems: A survey. arXiv. <https://doi.org/10.48550/arXiv.2411.07658>

